# Children First, Housing Later*

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**Abstract**

Should housing assistance require sobriety? Without externalities, no: if households are rational and forward looking, conditionality only removes an option. With externalities, conditionality fixes some externalities drug use imposes on users' children, but also exposes those it screens out to homelessness, whose externalities harm their children and everyone nearby. A general-equilibrium model with endogenous homelessness, drug use, drug prices and mortality, calibrated to United States data, captures those externalities, with altruism matching how much less parents use drugs than non-parents. Drugs are a luxury, so few use them, but because using risks early death, users are poorer. Full altruism cuts drug use and homelessness. Conditioning assistance cuts drug use in the long run, and, as excess revenues fund more vouchers for clean poor households, lowers homelessness slightly. Welfare rises in both the long run and the transition through raising children's lifetime resources, despite a temporary rise in childhood homelessness.

*Keywords:* homelessness; housing vouchers; substance use disorder; intergenerational spillovers; externalities.

*JEL classification:* I18; I38; H23; R21; J62.

**Pending updates from the new simulations**

transition: provisional, the second iteration of the new sixty-year run, which has not yet converged; every number and figure here is refreshed as it converges . . . . . 42

**Notes on this round's changes**

## 1 Introduction

The January 2024 point-in-time (PiT) count, run by the Department of Housing and Urban Development (HUD), found that at least 771,480 people in the United States were sleeping in homeless shelters or on the street, the most since the count began. In July 2025 an executive order moved the federal government decisively to one side of a long debate about what to do about it: it required sobriety in federally funded housing and homelessness programmes,[1] redirecting about a third of what the programmes at the time carried between them away from Housing First and towards programmes conditional on sobriety.[2] Whether housing assistance should be conditioned on sobriety is no longer an academic question. It is policy that decides the welfare of hundreds of thousands of people.

The disagreement between the Housing First and Treatment First camps is usually framed as a contest between two behavioural claims.[3] Critics of Housing First say that unconditional assistance creates moral hazard: guaranteeing housing assistance whatever a household does weakens the incentive to stay sober

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[1]Executive Order, *Ending Crime and Disorder on America's Streets*, 24 July 2025. Reported in `https://www.npr.org/2025/07/24/nx-s1-5479139/trump-homelessness-executive-order-civil-commitment-camping`.

[2]See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`.

[3]For the case against Housing First as the federal government now makes it, see the executive order cited above; for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`.

and attached to the labour market, so drug use and substance use disorder (SUD) rise, which feeds back to homelessness, because a SUD means worse employment prospects, which makes paying rent harder. In contrast, critics of Treatment First say that conditionality delays stable housing for exactly the households who need it most, worsens their employment outcomes, and makes SUD recovery harder, because the street and shelters are poor places to recover. That too runs in a circle: worse recovery means more disorder, and more disorder means more homelessness.

Randomised trials of Housing First find large and persistent gains in housing stability and mortality, and mixed-to-null effects on substance use (Tsemberis et al., 2004; Aubry et al., 2016; Costantini, 2025, among others). These trials, however, focus on the participants themselves, and run from six months to three years. As such, they cannot speak to the impacts either policy has on people who are not the participants or long-run participant behaviour and outcomes. A model can follow things that a trial cannot. It can trace the impact of a policy well beyond the people in the study group, from their children to others in the economy, and for far longer than the length of a trial.

This paper starts from the observation that the debate as framed is missing the mark. On its own, the moral hazard both sides argue about is not an inefficiency at all. Consider a model with no externalities, and households who are rational and forward looking, as in Becker and Murphy (1988), who do not have self-control problems of the kind modelled with hyperbolic discounting at the moment they decide whether to start using drugs. And suppose they account for the true hazard of acquiring a SUD when they make that decision, as the evidence suggests (Lundborg and Lindgren, 2002; van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014; Greenwood et al., 2025). Then, ceteris paribus, Housing First weakly dominates Treatment First for every household, because conditionality removes an option from an optimiser and gives it nothing back.

The case for Treatment First therefore has to rest on what drug use costs somebody other than the user, and this paper focuses on four such costs, externalities that randomised trials have not resolved: two that Housing First leaves in place and two that Treatment First creates.

The two Housing First leaves in place are harms drug use does to the user's children. The first runs through the household's budget, the *budget externality*: children live with their parents and share their consumption and their dwelling, but not the drug, so every unit a parent spends on drugs, and every unit of earnings it loses while a SUD keeps it unemployed, is consumption and housing the child goes without. When Treatment First reduces drug use, it diverts more resources towards children. The second runs through earnings in adulthood, the *earnings externality*: a child whose parent had a SUD during its upbringing starts adult life with lower productivity and earns less for the rest of its life. When Treatment First reduces the SUD rate, it raises the earnings of children in adulthood.

The two Treatment First creates both run through homelessness. By denying housing assistance to drug users, Treatment First exposes more of them to homelessness. The first externality this creates also falls on drug users' children, the *childhood homelessness externality*: a child lives wherever its parent lives, so a parent ending up in a shelter because of a sobriety programme it cannot comply with, or chooses not to, takes the child into the shelter too. The second falls on everyone, the *aggregate homelessness externality*: every household's utility falls with the share of the population that is homeless, capturing in a reduced-form way the idea that it is less safe and pleasant to live in a place with more homelessness.

In the model I design, a parent takes the budget, earnings and childhood homelessness externalities into account only through its imperfect altruism towards its child, and homeless households do not internalise at all the cost their homelessness imposes on everyone else. How much weight a parent puts on its child is calibrated to the difference in drug use between parents and adults without children in the data. In the 2024 National Survey on Drug Use and Health (NSDUH), adults without children use drugs 18.3 percent more than parents with the same characteristics, so the weight is set so that removing a parent's concern for its child would raise drug use in the model by that much. The part of the child's loss a parent does internalise is thus pinned down by observed behaviour.

Given the degree of altruism towards the child, the size of each externality is set by data. The budget externality depends on two things: how much a user household spends on the drug, and how likely it is to regain work when a SUD has left it unemployed. Both determine the resources left for the child. The weight given to drugs in the utility function is set to match the share of their income that drug users spend on drugs, 16.0 percent, from retail drug spending (Midgette et al., 2019) and the incomes of adults who use

in the 2024 NSDUH. How hard it is to find work again with a SUD is set to match the SUD share of the homeless, 21.7 percent (Gutwinski et al., 2021).

The earnings externality depends on the child's productivity penalty, which is set from a direct estimate: children exposed in utero to a Swedish alcohol experiment earn about 24 percent less as adults (Nilsson, 2017). The intergenerational transmission of SUD is then left untargeted: in the model a child of a parent with a SUD has 1.46 times the odds of acquiring one itself, against 1.44 to 2.51 in the data (Mellentin et al., 2016; Yoon et al., 2013).

The childhood homelessness externality also depends on two things, both of which shape what the child consumes: how much less housing a shelter provides than a dwelling on the market, and how much more likely a homeless parent is to be unemployed. The first is set so that the share of households who are homeless lies within 0.40 to 1.00 percent (HUD's 2023 PiT count and Burt et al., 2001). The second is set externally: a household in a shelter is 20 percent more likely to lose its job and 20 percent less likely to find one (Desmond and Gershenson, 2016). Finally, the aggregate homelessness externality is set to the annual willingness to pay for ending homelessness in the contingent valuation of Loubière et al. (2020), 0.20 percent of income.

Whose children bear these externalities depends on who uses drugs, and one feature of the model decides it. Two facts about drug use have to be reproduced together, and the literature has so far explained them only by assuming that lower-education households, whose income is lower because of their lower education, have a systematic preference for drugs (Greenwood et al., 2025). Drug use is concentrated at the bottom of the income distribution, where the average user's income is about 74 percent of the average in the 2024 NSDUH, and it is not universal: most households do not use drugs at all, so there is a large margin at zero, with a participation rate of 8.90 percent among adults aged 26 and over in the same survey.

To generate both I model drugs as a luxury good bought at a price, which puts most households at a corner where they optimally consume none. Left to its own devices, however, that same assumption concentrates use at the *top*, since the share of income spent on a luxury good rises with income. What reverses the income gradient, as I show, is the increased mortality risk that comes with a SUD: acquiring a disorder truncates the expected continuation value of life, and that loss rises with income, so in equilibrium use concentrates at the bottom. As far as I know, this is the first theory that can explain both why not everyone uses drugs and why drug users are poorer on average without resting on systematic differences in preferences.

I proceed with two models, a stylized model for the intuition and a full quantitative model for a comprehensive policy analysis. The stylized model is a life-cycle model with three periods of life (childhood, adulthood and old age) and rational forward-looking households, who care about consumption, housing and drugs. It shows four things at their clearest: how and why the income gradient in drug use reverses, that with no externalities Treatment First reduces welfare relative to Housing First, how one can find the optimal degree of drug use and SUD, and why the externalities Treatment First removes outweigh the ones it creates.

An adult in the model draws a productivity and an unemployment shock, and chooses how much of the drug to buy; drug users risk acquiring a SUD, and a SUD raises the chance of dying before old age and the chance of losing employment, and a job lost to the disorder is not covered by unemployment insurance. It chooses whether to live in a shelter or in a rental, and it may hold a means-tested voucher, which is rationed; food assistance puts a floor under its consumption, so that all households have a positive value of life, which means they have something valuable to lose by dying.

Children live with their parents during the parents' adulthood period and share their resources (except the drugs), so what a parent spends on drugs comes out of its child's consumption and housing, and the weight a parent puts on its child is calibrated to match the gap in drug use between parents and adults without children. The aggregate homelessness externality is in the model too, calibrated to the same willingness to pay as in the full model. At the end of the parents' adulthood, their child becomes an adult, and its productivity draw is penalised if the parent acquired a SUD while the child lived with them. There is no government budget in the stylized model, so revoked vouchers are not offset by a higher probability of drawing a voucher for sober households; that is deliberate, so the mechanisms can be seen clearly, and the quantitative model takes up the detail. Finally, I calibrate the stylized model to United States data.

Welfare is measured using the ex-ante expected utility of newborns, childhood included, in consumption-equivalent terms, against a world with rationed Housing First.

This stylized model shows starkly how the income gradient reverses. It separates the two penalties a disorder carries (death and unemployment). Scaled from nothing to 2 times its calibrated value, with the other penalty switched off, the death penalty takes the average user from 3.54 times average income to 0.52 times it; the unemployment penalty on its own leaves it at 3.54, against 3.54 with neither penalty.

The linchpin is whether a penalty can be insured against. The richer the household, the less a given quantity of the drug costs it in utility terms. Losing a job, on the other hand, does not raise the utility cost on the rich, because it can be insured against: savings grow with income, so a lost job takes about the same share of every household's resources, and in utility terms it costs a rich household no more than a poor one. No amount of job-loss risk can therefore flip the gradient. Dying is different, because no amount of savings can insure against it: it destroys the whole continuation value, including the value of being alive, and that value grows with income. The richer the household, the more the risk of dying outweighs the drug's lower utility price, so when the death penalty is large enough it is the poor who use rather than the rich.

The experiment is the reform itself. The programme stays at its benchmark rationing, with the same probability of drawing a voucher, and only the rule changes, from Housing First to Treatment First, which does not provide vouchers to households that use drugs or have a SUD. The stylized model keeps the price of drugs fixed; in the quantitative model it is endogenous.

The first result of the stylized model is the optimal level of drug use. To find it, I set the weight a parent puts on its child to one, so that nothing a parent does to its child is left external. In that economy nobody uses drugs, so nobody acquires a SUD, and homelessness drops by 3 percent.

The second result is that in the long run, without the externalities, 9.4 percent of households are strictly worse off under Treatment First and none is better off. However, counting what drug use costs the child, both the resources a parent diverts towards drugs and the earnings penalty of a parent's SUD, turns the ranking, from the first cohort of newborns on. This is because Treatment First lowers drug use, the SUD rate and, with them, childhood homelessness and aggregate homelessness, so in the long run it reduces all four externalities.

Treatment First lowers childhood homelessness by about 1 percent, because fewer households have a SUD and lose their jobs with no income support, and fewer are poor for having been born to a parent with the disorder. In the transition, the old and the adults alive at the reform lose 0.07 and 0.019 percent of consumption, while their children, the first cohort born into the reform, already gain 0.34 percent of lifetime consumption, because their parents are deterred while they live with them.

The full quantitative model that follows quantifies in finer detail the mechanisms and the welfare implications of Housing First against Treatment First, and settles whether the results of the stylized model survive in a richer environment that fits the data. It is a general-equilibrium life-cycle economy, and households here, as in the stylized model, care about consumption, housing and drugs. Children live with their parents and share consumption and housing resources with them while the parents are 26 to 50 years old, and become adults after that. Households choose whether to live in a rental or a shelter, and whether to use drugs. Using carries the risk of a SUD, from which a household recovers only with a small probability each year: recovery is exogenous, harder in a shelter and with serious mental illness (SMI), but more likely in treatment, which a household can choose at a cost. SMI arrives exogenously with age and is more likely for households with a SUD, and like a SUD it lowers employment and survival, all consistent with the data (Milby et al., 2005; Olfson et al., 2015; Nielsen et al., 2017; Murrie et al., 2020; Hjorthøj et al., 2023).

The price of drugs clears the drug market, so drug prices respond to policy, and in general equilibrium, as drug prices decline, using drugs becomes more appealing, which is a force pulling against the deterrence effect of Treatment First. The government budget balances through the probability of receiving a voucher, so the programme's reach responds to what it costs. Conditionality is enforced by a drug test that the government pays for but that misses some users; a household that is caught loses its voucher and is barred from the programme for a period of time. The quantitative model is calibrated to United States data as well, with a benchmark of rationed Housing First, in which about a quarter of eligible households get a voucher with no sobriety conditions.

The model is checked against various data that no parameter is fitted to, set out in Section 5.3. Among them: the age composition of the homeless, with the share of them employed and their income; the age profile of drug users, unemployment among those with a SUD, how much higher the drug use rate of households with SMI is relative to others', and excess drug-related deaths as a share of deaths between 25 and 44; the

share of renters spending more than thirty percent of income on housing, and the rent burden and wages of voucher holders; the income and net worth held at the very bottom of the distribution; the price elasticities of drug participation and of drug-use intensity; the odds that the child of a parent with a SUD acquires one, and that a homeless adult had a homeless parent; and how much higher three-year mortality is for households with SMI who lose their housing than for those who keep it (Costantini, 2025).

The quantitative experiments start in partial equilibrium: with the drug price and the probability of drawing a voucher held at their benchmark levels, Treatment First raises welfare relative to Housing First with the externalities and lowers it without them, as in the stylized model. In general equilibrium, all the stylized model's long-run and transition qualitative results, including welfare results, also carry over, except two. First, full altruism does not end drug use, but rather cuts it by 17 percent. Second, childhood homelessness does not fall in the long run or short run: it rises by up to 13 percent in the short run and remains unchanged in the long run.

# 2 Related literature

The literatures this paper draws on are set out in the order the argument needs them. The first three supply the machinery: what the trials establish about housing policy, how economists model the decision to use substances, and what general-equilibrium models of homelessness already deliver. The next two concern the externalities, which is where this paper departs from its predecessors: what a parent with a SUD transmits to their child, and how much weight a parent puts on the child who bears it. The last two are what the quantitative model is built out of: the hazards, taken one at a time from clinical and epidemiological work, and the drug market that makes the price endogenous.

**Randomised trials of Housing First.** A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public-service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol-related outcomes.

In the design of the model, I draw directly from Milby et al. (2005). Among cocaine-dependent homeless adults in day treatment, those given housing stayed abstinent longer than those given none, and those whose housing required abstinence did best. The model takes from this trial the penalty homelessness imposes on recovery.

Another relevant study, which I use to validate the model, is Costantini (2025). Using data on the programmes of the Department of Veterans Affairs, it finds that Housing First lowers the three-year mortality of homeless veterans with mental illness by 4.6 percentage points, while its Treatment First programmes, which combine short-term housing with treatment, have no lasting effect on mortality. The model produces outcomes consistent with these findings: among households with SMI, those whom a revoked voucher puts on the street are more likely to die within three years than those that keep their voucher, by an amount of the same order (Section 5.3).

A limitation of all of the randomised trials above is that they measure outcomes over six to thirty-six months, which is short against a recovery process that Vaillant (1995) documents as much more gradual and protracted: at the annual recovery rate the model takes from it for a housed household, 22 percent of those with a SUD recover within ten years. A further limitation of a trial, and perhaps a deeper one, is its scope. A trial randomises households within a fixed environment, and holds fixed exactly the things that decide the welfare comparison: the price of the drug, the government's budget, and, most importantly, the effects on children and other households that are not in the sample.

**Economic models of drug use.** Modelling substance use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find that the model-implied beliefs are actually very close to the true risk. Moreover, young people, if anything, overestimate the risk

of becoming dependent, and those who perceive a higher risk drink less (Lundborg and Lindgren, 2002). A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences. With such assumptions, Treatment First has a rationale even with no externalities at all. Therefore, that literature is complementary and cuts in the same direction: a present-biased household would welcome a sobriety requirement as a commitment device, so introducing it would strengthen the case for conditionality rather than weaken it.

Beyond the focus on externalities, this paper also departs materially from that literature in how it generates the income gradient in drug use. Greenwood et al. (2025) build a macroeconomic model of the opioid crisis in which households differ by education, and the concentration of use among the less educated comes about because of preference differences towards drugs across those groups. In contrast, in the model here no preference parameter varies with income: every household has the same weight on the drug, the same Stone-Geary endowment, the same hazard of acquiring a SUD and the same mortality penalty. The utility function does not change with the state the household is in, or with the choices it makes. The gradient comes out of the mortality channel alone, because a SUD destroys a continuation value and the continuation value rises with permanent income, so households with more to lose give up proportionally more than they gain from a drug that is cheap relative to their income.

**General-equilibrium models of homelessness.** I build directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental-health shocks, housing costs and housing policy (see also Abramson, 2026, on how eviction policy changes who becomes homeless). In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety-contingent policy has no behavioural margin to act on in their model. Adding endogenous use, disorder and recovery, together with state-dependent mortality, creates that margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival.

In this model the causation between the disorder and homelessness runs both ways, and the two directions reinforce each other. On the one hand, a SUD costs employment, which costs a household its housing. On the other hand, losing housing worsens employment outcomes and raises mortality, which raises the value of using relative to the value of life being risked, since households with less continuation left to lose risk less by using drugs. Each direction feeds the other, so what the model produces is a vicious circle rather than a one-directional flow.

**Intergenerational transmissions.** The earnings externality rests on three intergenerational links, and the literature documents each of them.

The first is that a parent's SUD changes the childhood. The administrative record of the child welfare system documents it at scale: parental drug abuse is recorded at more than a third of removals (AFCARS, FY2017). The epidemiological literature documents it too. Household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD have between one and a half and two and a half times the odds of acquiring one themselves when one parent is affected (Mellentin et al., 2016; Yoon et al., 2013).

The second link is that the childhood shows up in the child's own adult earnings, and the mechanism is as follows. A parent who remains unemployed longer because of its SUD, and who spends what money is left on the drug, has less of it for the child. Time investment in the child goes the same way. What the child loses is schooling, a good peer group, and the ordinary attention a household gives, in a home more chaotic than it would otherwise be. The child reaches the labour market with less human capital and fewer good connections.

The impacts on earnings have been measured. Dahl and Lochner (2012) find that family income raises children's achievement, and Restuccia and Urrutia (2004) put roughly half of the intergenerational correlation in earnings down to parental investment in education. Card (1999) puts the return to a year of schooling at around ten percent. And Nilsson (2017) closes the loop on a parental substance shock in particular: children exposed in utero to a Swedish experiment that sharply raised alcohol availability earn about twenty-four percent less as adults and finish less schooling. On the transmission of the disorder itself, Mellentin et al.

(2016) follow 398,881 offspring in the Danish registers and Yoon et al. (2013) a United States population-based sample, and both find that a child with one affected parent is more likely to acquire a SUD, at odds ratios of 1.44 and 2.51, respectively, which the model replicates.

What this paper adds is not another estimate but a price: it puts these intergenerational effects inside an equilibrium model where the parent's use decision is endogenous, so the transmitted loss becomes a quantity that policy moves rather than a coefficient.

The third link is that homelessness is persistent across generations as well. Earnings are persistent across generations: estimates of the intergenerational elasticity of earnings in the United States run from about four tenths (Solon, 1992) to about six tenths (Mazumder, 2005), and Chetty et al. (2014) find the same persistence in ranks across the whole distribution. The model carries this persistence from parent to child, and that channel alone is enough to generate, inside the model, the higher likelihood observed in the data that children of homeless parents are homeless themselves (Bassuk et al., 1997): a child of a homeless parent tends to draw a lower productivity because its parent's was low, not because of the homelessness itself.

There is evidence that a poor environment lowers a child's earnings beyond what the persistence of skills across generations explains, although it concerns poor neighbourhoods rather than homelessness. Chyn (2018) finds that children displaced by Chicago's public-housing demolitions earned sixteen percent more a year as adults. Chetty et al. (2016) estimate that a move to a lower-poverty neighbourhood before age thirteen raises adult earnings by about a third, and the gain falls with the age at the move, which says that what is transmitted is the environment a child is raised in rather than anything the child carries.

The model imposes no earnings penalty for growing up in a shelter, but adding one would not change its conclusions: under Treatment First the share of children in shelters does not rise in the long run, while the share of children living with a parent who has a SUD falls by 1.92 points, and even in the short run the absolute increase in childhood homelessness is about 35 times smaller than the mass of children spared a parent who uses drugs.

**Parental altruism towards children.** The extent to which externalities exist depends on the degree to which the parent does not internalise them, that is, one minus the weight it puts on its child, and that weight has a literature of its own. Altonji et al. (1997) test perfect altruism on the money parents give their adult children, using survey data that record the incomes of both parent and child. Among families in which parents give to their children, they estimate how much more parents give when their own income is a dollar higher, and how much more when their child's income is a dollar lower. Moving a dollar from the child to the parent does both at once and leaves the family's total income unchanged, so the two together say what that move does to the gift. Perfect altruism says the parent would give the whole dollar back; they find at most thirteen cents, and the calibrated model implies a low degree of altruism as well. Parents are probably altruistic, but not completely, which is the case for treating that weight as strictly between zero and one rather than assuming either end of it.

The Altonji et al. experiment cannot be run here. It needs a parent and a grown child with budgets of their own at the same time, linked by the parent's transfers. In the model, the child shares its parent's budget while the parent is 26 to 50 years old, and the model does not track the relationship between the two after the child becomes an adult. What the model can do is another test, and for this paper it is a more relevant one: it compares parents with adults without children. In NSDUH 2024 past-month use among non-parents is 18.3 percent above use among parents, reweighted to the non-parents' joint distribution of age, income, sex and marital status so that the gap is behaviour rather than composition or observed characteristics. The weight on the child's utility in the household's problem is the only thing in the model that makes a household use less than it otherwise would, so the weight is set so that setting it to zero (mimicking having no children) raises drug use by the same 18.3 percent.[4] The externality is therefore not assumed to be large; its size is calibrated to observed behaviour on exactly the margin this paper is about.

**What the hazards are built from.** The model's transitions are not free parameters dressed as structure; each is anchored on, or checked against, a clinical or epidemiological estimate, and the calibration section

[4]Drug users are nearly as likely as anyone to have children: in the National Survey of Family Growth 2017–2019, 75 percent of men and women aged 40 to 49 who used cocaine, crack or methamphetamine or injected a drug in the past year have had a child, against 81 percent of those who used none.

discusses them one at a time. Four groups do the work. On mortality, the excess death rate of a SUD is calibrated inside the model, to the average income of users, since it is what decides how far down the income distribution drug use sits. The rate it lands on, 4.00 times the baseline mortality risk, is in line with the estimates in the literature: Plana-Ripoll et al. (2019) put the mortality rate ratio of a SUD at 3.91 in the Danish registers, and Heiberg et al. (2018) at 6.6 among Norwegians in specialist care for one. Meyer et al. (2023) put the death rate of homeless adults at about 1.60 times that of housed adults who are also poor, and the model applies that ratio to homeless households, and Olfson et al. (2015) measure the excess death rate households with SMI face; against those hazards the value of a life comes from the willingness-to-pay estimates surveyed in Viscusi and Aldy (2003).

On the SMI chain, Nielsen et al. (2017) identify the effect of a SUD on the onset of SMI in Danish registers, Gutwinski et al. (2021) meta-analyse its prevalence among the homeless, and Bradizza et al. (2006) review relapse under dual diagnosis.

On recovery, Vaillant (1995) documents how slow unassisted remission is, and Milby et al. (2005) supply a causal estimate: a randomised trial on cocaine-dependent homeless adults in which the arm housed without conditions was abstinent 6.60 weeks against 4.48 in the arm given no housing, which is where the penalty homelessness imposes on recovery comes from.

On employment and the cost of housing loss, Desmond and Gershenson (2016) estimate what losing housing does to holding a job, a penalty the model also applies to finding one, and Loubière et al. (2020) price the disamenity of visible homelessness through a contingent valuation, which is what makes the aggregate headcount enter anybody's utility at all.

**The drug market.** Making the drug price endogenous is what lets a policy that changes the number of users change what using costs everybody else. Three elasticities govern it and they do different jobs, so the literatures behind them are worth separating: the demand side is driven by how much the extensive and the intensive margins respond to changes in price, and the supply side determines how much the equilibrium price moves when demand changes.

*The price elasticity of demand.* The demand side has two margins and the model produces both. The first is the extensive margin (participation), which is whether a household uses at all. The second is the intensive margin, how much a user buys once using. Neither is calibrated. Both are produced by the model and checked against the literature in Section 5.3.

Both margins have been measured empirically, and between them the estimates bound how much a policy can reduce use once the drug price falls: if both elasticities are low, a decline in use that lowers the price does not bring many users back. Estimates of the participation elasticity run from about $-0.3$ to $-0.9$ (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998). The model's participation elasticity, $-1.32$, is somewhat larger in absolute value than these estimates, so in the model a fall in the drug price brings back more users than it would in the data. That works against Treatment First: as it deters use, the drop in price brings back more users than the evidence suggests, so the model if anything understates how much the policy reduces drug use, and the welfare gain that comes from lower use.

The literature estimates the intensive-margin elasticity to be of a similar magnitude to the extensive-margin elasticity, between about $-0.3$ (Grossman and Chaloupka, 1998) and $-0.8$ (Olmstead et al., 2015). The model's intensive elasticity, measured over all users, $-0.63$, is in the middle of these estimates, so the model matches how much each user's purchases respond to the drug price.

A separate point about rational drug use follows from the findings of all these studies: participation and intensity both respond to price, even for drugs as habit-forming as heroin and cocaine, and long-run responses are consistently larger than short-run ones. That ordering is the signature of forward-looking behaviour rather than myopia, which is why it belongs with Becker and Murphy (1988) above as well as here.

*The elasticity of supply.* This is the general equilibrium offset, and it runs against Treatment First in particular: deterring households from using takes them out of the market, which causes demand and the drug price to fall, but then a cheaper drug draws others in. I am not aware of a direct estimate for an illicit drug. Cannabis is the closest market I found to an illicit drug, as it remains a Schedule I substance federally, so even a licensed producer works without interstate trade, without ordinary banking and under

federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al. (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the elasticity the model uses, which is at the low end of what the literature uses: where studies of illicit markets take a position, they treat supply as very elastic, because the drugs are cheap and simple to produce (Rhodes et al., 2002).

**In a nutshell.** The trials establish that housing stability, SUD recovery and mortality respond to policy but cannot price what happens in the long run, to others not in the sample, or to the budget that pays for it. The rational addiction literature models the use decision but abstracts from housing, children, and SMI. The general-equilibrium homelessness literature delivers a housing market equilibrium but has no drug margin for conditionality to act on. The transmission literature measures what having a parent with a SUD does to a child but takes the parent's behaviour as given. This paper's contribution is to put them in one framework and then to insist on a distinction none of them needs on its own: that the moral hazard which dominates the policy debate is not a reason for conditionality, and that whatever case exists has to be made on the externalities.

# 3 A stylized model

## 3.1 Three questions

In the data, drug use is concentrated among low-income households, most of the homeless do not carry a SUD, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has seven state variables and fifty-nine parameters, which is what the welfare and fiscal questions require for a satisfactory analysis, but it also makes it hard to see which pieces of the environment do the heavy lifting. To help see the main drivers in the model, this section strips the environment down to the basics, but only to the point where the main questions of this paper are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They know the true hazard of acquiring a SUD, they are aware of what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, myopia, self-control, or on a taste that changes once the household has developed a SUD. The model abstracts from government and from how the programmes are funded; both enter the quantitative model of Section 4. The stylized model answers three main questions.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution, and it is not universal there either: most poor households do not use at all. Both facts have to hold together. I model drugs as a *luxury* good bought at a price, which is what delivers the second: not all households choose to consume drugs. However, left to itself the Stone-Geary form in equation (2) below means the share of income spent on drugs rises with income, so when the environment is calibrated to the use rate, use is concentrated at the *top*. Two risks push the other way, and each costs the household something: a SUD raises the chance of unemployment, which costs earnings, and also raises the chance of dying, which costs the value of the life it has left. The question is which of the two reverses the concentration from the top to the bottom, and I show that it is the risk of death.
2. **How does housing assistance shape the drug-use decision?** Housing assistance pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: I refer to it as the value-of-life force, and it lowers use. In contrast, if the voucher is unconditional, then it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: I call this the moral-hazard force, and it raises use. Both are first order, so theory does not sign the net effect of unconditional housing assistance on drug use, and the question is quantitative.
3. **When does Treatment First improve welfare relative to Housing First?** In the stylized as well as the full model, without externalities, never. Housing First always raises welfare when agents are forward looking, as they are in the models in this paper. Unconditional assistance creates moral hazard through its insurance value, but that alone means Treatment First can only lower welfare, because it

only removes an option that a household who does not have self-control issues can commit to taking under Housing First as well. However, if parents do not fully weigh their children's welfare in their decisions, then drug use and SUD generate externalities, such as the budget, earnings and childhood homelessness externalities, and Treatment First may be preferred. By setting the weight parents give to their children's utility in their own objective function, the model can also say what the optimal drug use and SUD rates are.

The model is built in stages, and each is recalibrated before it is used, so every result below is read off a model that is fitted closely to its targets. The first stage has no child living with its parents and no externalities: nothing a household does lands on anyone else. It answers the first two questions and shows that Treatment First cannot raise welfare. The second adds the child, who lives in the household during the parent's adulthood period, shares its consumption and housing, and inherits a lower lifetime productivity if the parent acquired a SUD in that period. This stage brings in the budget, the earnings and the childhood homelessness externalities, shows what drug use would be if parents counted their children fully, and compares Treatment First with Housing First, in the long run and along the transition to it. The third adds the aggregate homelessness externality, the disamenity every household suffers due to the homelessness of others around it, and shows the results remain essentially unchanged with that addition.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and components that obscure the mechanisms have been removed with the model refitted without them. What remains are the main drivers of the results. The welfare evaluations below are for illustration and intuition only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits many aspects of the data.

## 3.2 Stage 1: households on their own

### 3.2.1 Households and preferences

In stage 1, without children, there are two periods (adulthood and old age) and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty-six to eighty-five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is $\beta$ per period and $\bar{\phi}$ is survival over thirty years. Saving earns no interest ($R = 1$), so a unit saved is a unit carried into the next period. Notation follows the full model in Section 4 wherever the two share an object, except that employment and the SUD are coded more simply here, as set out below.

A household is characterised by its productivity $y$, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Labour is supplied inelastically. Households differ in $y$, in age $j \in \{1, 2\}$, in employment status $e \in \{-1, 0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility does not count the child, absent in this version of the model. Non-durable consumption $c$, housing services $h$ and drug consumption $v$ form the bundle

$$X = \left[c^{1-\gamma} h^{\gamma}\right]^{1-\theta} (b + v)^{\theta}, \tag{1}$$

and flow utility is

$$u(c, h, v) \;=\; U(X) \;+\; \aleph, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{2}$$

The weight on housing $\gamma$ is its share of the non-drug bundle, the weight on the drug $\theta$ is its share of the whole, and $\sigma$ is the coefficient of relative risk aversion. For simplicity, drugs can be chosen only in the first period, and housing is chosen each period from a three-point grid $\{h_1, h_2, h_3\}$. The pair $(\theta, b)$ governs the drug term: the marginal utility of the $v$th unit is proportional to $\theta/(b + v)$, so $\theta$ mostly sets how much a user uses and $b$ whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non-users at all, because without the shifter every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish even when consumption is low. Households can save in one risk-free asset, denoted as $a$, and cannot borrow.

### 3.2.2 Timing

**Period 1.** Every household enters the period clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability $u_0$, and a household that is income eligible draws a voucher with probability $\pi_V$; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$ and saving, paying market rent $p_r$ on whatever unit it picks. It starts with no assets.

Using drugs carries the risk of a SUD, the same whatever the quantity,

$$\pi(v) \;=\; \pi_d \, \mathbf{1}\{v > 0\}. \tag{3}$$

The whole risk therefore sits on the decision to use at all, and $\theta$ alone sets how much a user uses once it has decided to actually use.

**Acquiring a SUD and the death penalty.** The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probabilities

$$\phi(0) \;=\; \bar{\phi}, \qquad \phi(1) \;=\; \bar{\phi}^{\,\kappa_{\phi d}}, \tag{4}$$

where a SUD raises the probability of surviving the period to the power of $\kappa_{\phi d} > 1$, thus lowering it. The mortality penalty applies here, and here only.

**Period 2.** This is the terminal period, after which the household dies with certainty. A household without a SUD is unemployed with probability $u_0$ and one with a SUD with the higher probability $u_1$; if unemployed it receives unemployment insurance (UI), unless the disorder is what cost it the job (Section 3.2.3). The voucher is drawn again, subject to the policy rule (Section 3.5). The household chooses consumption and housing, there is no drug-use decision, and assets are run down with nothing bequeathed.

### 3.2.3 Employment, income support and housing

A SUD costs a household resources in one way only: it raises the chance of losing its job from $u_0$ to $u_1$, and UI does not provide income support for the extra job losses. A household with a SUD loses its job for the ordinary reasons anyone does, with probability $u_0$, and because of the disorder with the further probability $u_1 - u_0$. Employment status $e$ therefore takes three values: working ($e = 1$), out of work and covered by UI ($e = 0$), and out of work because of the disorder and not covered by UI ($e = -1$). Income in a period, given permanent productivity $y$ and employment status $e$, is

$$y^g(y, e) \;=\; \begin{cases} y & \text{if } e = 1, \\ \psi_{ui} \cdot \min\{y, \bar{y}\} & \text{if } e = 0, \\ 0 & \text{if } e = -1, \end{cases} \tag{5}$$

with $\psi_{ui}$ the replacement rate and $\bar{y}$ a cap on the earnings base, and $e = -1$ occurs with probability $u_1 - u_0$ for a household with a SUD and zero for every other. There is no other income risk.

Food assistance is paid in kind. A household with income $y^g$ in a period receives

$$F(y^g) \;=\; \max\{0,\; \bar{c} - \varrho \, y^g\} \tag{6}$$

of consumption, which it cannot spend on rent or on drugs, so its other income has to cover those. Food assistance is essential for guaranteeing that every household has a positive value of life: it puts a floor under the consumption of every household, including one that loses its job to a SUD, so that every household has something valuable to lose by dying.

Each period the household chooses

$$h_k \in \{h_1,\; h_2,\; h_3\}, \qquad h_1 < h_2 < h_3, \tag{7}$$

where $h_k$ can be a shelter ($h_1$), a small unit ($h_2$) or a standard unit ($h_3$). A shelter is free and delivers fewer housing services than the market does.

Vouchers are rationed and means-tested, and when provided they cover part of the rent. A voucher is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le \kappa_V\, y^{\text{med}}\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{8}$$

where $\kappa_V\, y^{\text{med}}$ is the threshold for eligibility, $y^{\text{med}}$ is median income, and a household holding one pays the fraction $\kappa_r$ of market rent, so the rent paid on housing option $k$ is

$$r_k(\eta) \;=\; p_r\, h_k\big[\kappa_r \eta + (1-\eta)\big] \cdot \mathbf{1}\{k \neq 1\}, \tag{9}$$

which is zero when the household is homeless. Vouchers are provided in both periods of life, and $\pi_V$, the share of eligible households receiving vouchers, is a policy dial of Section 3.5.

The two policies differ in exactly one respect, the last indicator in equation (8). **Housing First** does not condition vouchers on sobriety, and thus does not revoke them from income-eligible households who consume drugs ($v > 0$) or have a SUD ($d' = 1$). **Treatment First** revokes vouchers from income-eligible households unless they satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. In the stylized model, detection of drug use and SUD is assumed to be perfect, an assumption I relax in the full model. The condition is applied to the period in which the voucher is paid, so what it tests differs by period: every household is born clean, so $d' = 0$ holds for everyone in period 1 and only $v$ can fail the condition in that period, while in period 2 there is no drug choice, $v = 0$ holds by construction, and only the disorder, $d' = 1$, can fail it.

A voucher's revocation in period 1 is a function of the choice $v > 0$ in the same period, so the timing is worth stating exactly. Every household arrives clean, and the period-1 voucher is drawn before $v$ is chosen. This means that conditionality cannot screen anyone out at entry. The order within the period is therefore: the voucher is drawn, the household chooses $v$, the policy rule is then applied to the voucher it drew, and it chooses housing knowing whether or not its voucher was revoked and the effective rent it faces. So under Treatment First a household that uses (period 1), or carries the disorder (period 2), is subject to full market rent immediately, in the period concerned.

### 3.2.4 The household's problem

Let $a$ be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$V_2(a', y, e', \eta', d') = \max_{\{k\}} \Big\{ u(c, h_k, 0) \Big\}, \qquad c = Ra' + y^g(y, e') - r_k(\eta') \;+\; F\big(y^g(y, e')\big) \;\ge\; F\big(y^g(y, e')\big), \tag{10}$$

where consumption cannot fall below the food assistance, which is paid in kind. I write $V_2(a', y, d') = \mathbb{E}\big[V_2(a', y, e', \eta', d') \mid y, d'\big]$ for its expectation over the period 2 employment and voucher draws, the SUD having already been realised at the transition. In period 1 the household draws $(e, \eta)$ and then chooses housing, drug use and saving,

$$V_1(y, e, \eta) = \max_{\{k,\, v \ge 0,\, a' \ge 0\}} \Big\{ u(c, h_k, v) + \beta\, \mathbb{E}_{d'} \Big[ \phi(d')\, V_2(a', y, d') \Big] \Big\}, \tag{11}$$

with $c = y^g(y, e) - r_k(\eta) - p_v v - a' \;+\; F\big(y^g(y, e)\big) \;\ge\; F\big(y^g(y, e)\big)$, and $d' = 1$ with the probability of (3). Survival sits inside the expectation over $d'$ because it depends on which $d'$ is realised. A household uses if the value at its best positive $v$ exceeds the value at $v = 0$.

## 3.3 Calibration

Each stage is fitted to the same targets. Table 1 lists what is set outside the model and Table 2 what is fitted inside it, with a column for each stage: stage 1 of this section, and stage 2, with the child that Section 3.7 adds, which is refitted with the child in place. The fit uses the quantitative model's rule, moving all internal parameters together, each in proportion to the gap on its own moment until each gap is within a certain degree of tolerance.

Every target is documented in full in the replication package's data appendix, `data/DATA_APPENDIX.md`, which names the dataset behind each one, its year and universe, and the computation that turns it into the number quoted here. What follows gives the moment and its source; the appendix gives the arithmetic.

### 3.3.1 External calibration

Preferences and demographics are conventional: the discount factor over a thirty-year period is $\beta = 0.86$, which is 0.995 a year compounded,[5] risk aversion is $\sigma = 2$, as in the quantitative model, and survival over a period, the cumulative chance of surviving from age twenty-six to age fifty-five, is $\bar{\phi} = 0.93$, from the life table for the total population of the United States in 2010 (Arias, 2014).

The chance that drug use causes a SUD, $\pi_d$, is also externally calibrated. This is because drug use and the SUD are fully separated by cohort: the young are the users, and the old just carry the SUD. So the SUD rate is participation times $\pi_d$, and $\pi_d$ is the ratio of the two in NSDUH 2024, 3.68 percent of adults aged 26 and over carrying a moderate or severe SUD against 8.90 percent carrying one of any severity, the measure of use the model matches, which gives $\pi_d = 0.41$.

UI replaces half of previous earnings on a base capped at mean income, in line with statutory state rules, and nothing when a job is lost to the disorder, which those rules treat as misconduct (U.S. Department of Labor, 2024). Food assistance tapers at $\varrho = 0.30$, the benefit reduction rate of the Supplemental Nutrition Assistance Program (SNAP). The rent per unit of housing services $p_r$ and the price of a unit of drugs $p_v$ are normalised to one and held fixed in every counterfactual. Both rental units are set, the small one at $h_2 = 0.10$ and the standard one at $h_3 = 0.40$ of mean income. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, a simplification of the HUD rule, under which a tenant pays thirty percent of its income and the voucher pays the rest of the rent only up to the fair market rent (the latter condition being abstracted from); eligibility is capped at half of median income, $\kappa_V = 0.50$, and $\pi_V = 0.27$ is the ratio of recipients to the income-eligible population in the 2023 Picture of Subsidized Households (PSH), published by HUD.

### 3.3.2 Internal calibration

The dispersion of log permanent income $\sigma_y$ is set to match how unequal earnings are at the bottom: in the 2022 Survey of Consumer Finances (SCF), households aged 26 to 65 with positive wage and salary income hold 17.2 percent of total earnings in the poorer half of the distribution. The weight on housing $\gamma$ is picked to match the 26.8 percent of non-drug expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX). The flow value of being alive $\aleph$ is matched to a value of a statistical life (VSL) of 80 times average annual income, about \$9.0 million against a mean household income of \$113,000, which is within the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars.

The parameter $b$, which sets the marginal utility of the first unit of drugs, is chosen to match a drug participation rate of 8.90 percent from the 2024 NSDUH for respondents aged 26 and over with a disorder of any severity, and the weight on drugs $\theta$ is set to match the aggregate drug spending over the aggregate income of users, 16.0 percent, from retail spending in Midgette et al. (2019) and the incomes of users in the 2024 NSDUH. The level of food assistance $\bar{c}$ is fitted so that total food assistance is 0.42 percent of total labour income, the ratio of SNAP outlays to gross domestic product (GDP) in the data.

The mortality rate ratio $\kappa_{\phi d}$ is set to match the income of users, 0.74 of the population average in the 2024 NSDUH, because the death risk is what decides how far down the income distribution use sits, as Section 3.4 shows. Housing services in a shelter $h_1$ are chosen so that the homelessness rate in the model is 0.60 percent, inside a band of 0.40 to 1.00 percent of households. The floor is HUD's 2023 PiT count, a snapshot of a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). The unemployment probability without the disorder $u_0$ is chosen such that the aggregate unemployment rate is 4.85 percent, the average over 2014 to 2024 in the Bureau of Labor Statistics (BLS) Current Population Survey (CPS), and the unemployment probability with the disorder $u_1$ is set such that the share of the homeless carrying a SUD is 21.7 percent, consistent with the meta-analysis of Gutwinski et al. (2021).

[5]The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon.

Table 1: Externally set parameters

| Parameter | Description | Source or rationale | Value |
|---|---|---|---|
| *Preferences and demographics* | | | |
| $\beta$ | Discount factor over a thirty-year period | 0.995 annually, compounded | 0.86 |
| $R$ | Gross return on saving | Zero percent interest | 1.00 |
| $\sigma$ | Risk aversion | Standard; Attanasio and Weber (2010) | 2 |
| $\bar{\phi}$ | Survival over a period, no SUD | US life table 2010, Arias (2014), ages 26–55 | 0.93 |
| *The SUD* | | | |
| $\pi_d$ | Chance that use starts a SUD | NSDUH 2024: SUD rate over participation | 0.41 |
| *Income* | | | |
| $\psi_{ui}$ | UI replacement rate | Department of Labor, statutory rates | 0.50 |
| $\bar{y}$ | Cap on the earnings base | Department of Labor, state benefit maxima | 1.00 |
| $\varrho$ | Food assistance taper | SNAP benefit reduction rate | 0.30 |
| *Housing, drugs and the voucher* | | | |
| $p_r$ | Rent per unit of housing services | Normalisation | 1.00 |
| $p_v$ | Price of a unit of drugs | Normalisation | 1.00 |
| $h_2$ | Small rental unit | Externally set | 0.10 |
| $h_3$ | Standard rental unit | Externally set | 0.40 |
| $\kappa_r$ | Tenant's share of the rent | HUD voucher rules | 0.30 |
| $\kappa_V$ | Voucher means test | HUD income limits | 0.50 |
| $\pi_V$ | Chance of drawing a voucher | HUD 2023, recipients over eligible | 0.27 |
| *The child, stage 2 onward* | | | |
| $\xi$ | Equivalence scale while the child is at home | Square-root scale | 1.41 |
| $\rho$ | Persistence of productivity across generations | Solon (1992); Mazumder (2005) | 0.50 |
| $\delta_d$ | Child's productivity penalty, parent had a SUD | Nilsson (2017), a 24 percent cut | 0.27 |

Notes: stage 1 is the model of Section 3.2, without the child; the parameters of the child enter at stage 2, in Section 3.7.

Table 2: Internally calibrated parameters, by stage

| | | | Stage 1 | | Stage 2 | |
|---|---|---|---|---|---|---|
| Parameter | Target moment | Data | Model | Value | Model | Value |
| $\sigma_y$ | Bottom half's share of earnings, SCF 2022 | 17.2% | 21.1% | 0.99 | 21.1% | 0.99 |
| $\gamma$ | Housing over non-drug expenditure, CEX 2024 | 26.8% | 26.8% | 0.40 | 26.8% | 0.40 |
| $\aleph$ | VSL over income | 80 | 84.0 | 11.25 | 80.0 | 13.31 |
| $b$ | Drug participation, NSDUH 2024 | 8.9% | 8.9% | 0.054 | 8.8% | 0.053 |
| $\theta$ | Drug spending over users' income | 16.0% | 15.8% | 0.19 | 16.0% | 0.18 |
| $\kappa_{\phi d}$ | Users' income over the average, NSDUH 2024 | 0.74 | 0.75 | 2.58 | 0.73 | 2.68 |
| $h_1$ | Homelessness rate | 0.6% | 0.6% | 0.041 | 0.6% | 0.042 |
| $u_0$ | Unemployment rate, BLS CPS 2014–24 | 4.9% | 4.8% | 0.046 | 4.9% | 0.047 |
| $u_1$ | SUD share of the homeless, Gutwinski et al. (2021) | 21.7% | 22.0% | 0.100 | 21.6% | 0.10 |
| $\bar{c}$ | Food stamp spending over labour income | 0.42% | 0.42% | 0.11 | 0.42% | 0.11 |
| $\omega_c$ | Use of non-parents over parents, less one, NSDUH 2024 | 18.3% | – | – | 18.2% | 0.0021 |

Notes: stage 1 is the model of Section 3.2, without the child; stage 2 adds the child of Section 3.7 and refits every parameter; $\omega_c$ exists only once the child does.

## 3.4 Who uses drugs

With the calibrated stage 1 model in hand, this section asks which feature of the model puts drug use at the bottom of the income distribution. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: in the baseline of stage 1 the average user then earns 3.54 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel*: the SUD increases the chance of dying, through $\kappa_{\phi d}$ in equation (4). The *unemployment channel*: a SUD costs the household a job with additional probability $u_1 - u_0$, a job loss that UI does not cover, and if its savings and food assistance cannot cover the rent then the household downsizes, consumes less, or loses its home for a shelter, which is free but provides housing services that are below those of the small rental.

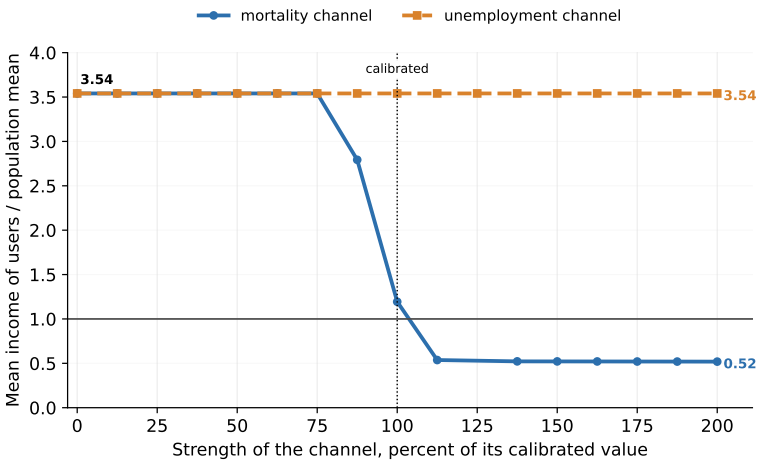



Figure 1: What it takes to reverse the income gradient in drug use

Notes: stage 1, the model without the child. Each curve scales one channel with the other switched off, so both start from the economy with no consequences of a SUD. The vertical axis is average income of users over the population average, and $b$ is recalibrated at every point so that participation is the calibrated 8.95 percent. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

Figure 1 scales each channel on its own, with the other switched off, from nothing to 200 percent of its calibrated value. Scaling the mortality channel multiplies $\kappa_{\phi d}$; scaling the unemployment channel moves $u_1$ up from $u_0$, reaching its calibrated value at 100 percent and continuing all the way up to 200 percent of its calibrated value. Apart from the two channels, every parameter stays at its calibrated value except $b$, which is recalibrated at every point so that participation stays at the calibrated 8.95 percent: the amount of drug use is the same everywhere on the figure, and only who uses changes. Both curves therefore start from the same economy, the one with no consequences of a SUD at all.

On its own, the mortality channel takes the income ratio from 3.54 with nothing switched on down to 1.19 at its calibrated strength, and falls further to 0.52 as it continues to rise to 200 percent of its calibrated value. With both channels at their calibrated strength the ratio is 0.75: the unemployment channel, which does nothing on its own, adds to the mortality channel once it is in place. What mortality destroys is a continuation value, and a household only stops using when the expected loss of that value exceeds what the drug is worth to it.

In contrast, the unemployment channel does not reverse the gradient at all: on its own it leaves users at 3.54 times average income at its calibrated strength, against 3.54 with nothing switched on, and at 200 percent they are still at 3.54. A resource penalty does not produce the concentration of drug use at the bottom.

Why should the two channels differ at all, when both punish the household? Because the cost of only one of them rises quickly enough with the household's income. Who uses is decided by a race between two things that change with income. On one side is the utility cost of the drug: a household with a higher income gives up less marginal utility for each dollar it spends on drugs, so buying $v$ units costs a rich household little in terms of utility relative to a poor household.

On the other side are the expected penalties from acquiring a SUD. The unemployment penalty cannot offset the falling utility price of the drug, because it does not rise with income. A job the disorder costs is not covered by UI, but the household can save, and savings grow with income, so a richer household that loses its job to a SUD keeps about the same share of its resources as a poorer one. Since losing the same share costs a rich household no more in utility terms than a poor one, the penalty weighs on the rich no more than on the poor. Raising it reduces use, but once $b$ brings the drug use rate back to its target, the drug is still cheapest for the rich and use stays at the top.

Mortality is a different object. Dying is a disproportionate loss that cannot be mitigated through insur-

Table 3: Stage 1: the two forces and the welfare of conditionality

| | None | Treatment First | Housing First |
|---|---|---|---|
| *Panel A. Outcomes, percent of households* | | | |
| Drug use | 7.61 | 4.45 | 8.95 |
| Carrying a SUD | 3.15 | 1.84 | 3.70 |
| Homeless | 0.79 | 0.56 | 0.59 |
| *Panel B. The two forces on participation, percentage points* | | | |
| Value of life, Treatment First less none | | −3.16 | |
| Moral hazard, Housing First less Treatment First | | | 4.50 |
| *Panel C. Treatment First against Housing First* | | | |
| Households strictly worse off, percent | | 9.4 | |
| Households strictly better off, percent | | 0.0 | |
| CEV of those affected, percent | | −0.19 | |
| CEV of the population, percent | | −0.017 | |

Notes: stage 1, the model without the child and without the disamenity of homelessness. None is no programme. Welfare changes are consumption-equivalent variations (CEV): the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value, negative when Treatment First reduces welfare relative to Housing First.

ance. It destroys the whole continuation value: the consumption of a life still to be lived, and $\aleph$, the value of being alive at all, which no amount of savings can protect. A richer household, therefore, has more to lose by dying, resulting in an expected penalty to using that rises with income. This means that once income is high enough the cost outweighs what the drug is worth to the household, and at that point it is the rich who stop first.

## 3.5 How does housing assistance shape the drug-use decision?

The two forces of question 2, moral hazard and the value of life, can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value-of-life force operates. Under Housing First a voucher is not revoked for sobriety-related reasons, so both the moral-hazard and the value-of-life forces operate. Comparing the two policies at the same generosity therefore isolates the moral-hazard force, and comparing Treatment First with no programme at all isolates the value-of-life force.

Panels A and B of Table 3 put numbers on both, in the baseline of stage 1. With no programme at all, 7.61 percent of households use. Treatment First takes that down to 4.45 percent, which means that the value-of-life force reduces drug use by 3.16 percentage points. Switching to Housing First then raises it above what it was before a programme existed, to 8.95 percent, so the moral-hazard force raises drug use by 4.50 percentage points and the net effect of both forces under Housing First is a 1.33 percentage-point increase in drug use relative to no vouchers at all.[6] Both comparisons here are between long-run steady states, each the rest point of the economy under a policy that has always been in place. Both forces are first order and they work against each other, which is the central feature of the problem. The point of this analysis, beyond showing how to isolate these forces, is to illustrate that anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

## 3.6 Welfare

### 3.6.1 Without externalities, Housing First always wins

Sections 3.4 and 3.5 established what each policy does to behaviour. This section asks what each does to welfare, comparing the two with the same $\pi_V$ and $p_v$, in the baseline of stage 1, where nothing a household does affects anyone else. Welfare is measured throughout as a consumption-equivalent variation (CEV), set

[6]The SUD rate follows participation one for one, since every user runs the same risk: 3.15 percent of households carry the disorder with no programme, 1.84 under Treatment First and 3.70 under Housing First. Homelessness falls under both programmes, from 0.79 percent to 0.56 under Treatment First and to 0.59 under Housing First.

out formally in Section 6.1. The policies differ only in the last indicator in equation (8), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a rational household with no commitment problems cannot be worse off under Housing First:

$$V_1(y;\ \text{Housing First},\ \pi_V)\ \geq\ V_1(y;\ \text{Treatment First},\ \pi_V) \qquad \text{for every } y, \tag{12}$$

with equality exactly for households that would not use under Housing First for any combination of unemployment and voucher draws they may receive in period 1. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward-looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.

It follows that the moral hazard is not an inefficiency. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household taking a risk it rationally chooses to bear. The moral-hazard force of Section 3.5 is a statement about drug use and not about welfare.

Panel C of Table 3 puts welfare numbers on it: 9.4 percent of households are strictly worse off under Treatment First,[7] none is better off, and those affected lose the equivalent of 0.19 percent of consumption. Those are the households that would use under Housing First after some combination of draws, and so could lose a voucher under Treatment First, in period 1 or, if they acquire a SUD, in period 2. A household that would never use, regardless of its unemployment and voucher draws, is indifferent to which programme is in place.

Therefore, the case for conditionality in this model has to come from elsewhere: from what a household's drug use does to somebody else.[8]

## 3.7 Stage 2: the child

Up until this point, I have shown that without externalities, Housing First weakly dominates Treatment First. From a model point of view, stage 1 assumed that the child only appeared as an adult: nothing its parent did affected it, and nobody counted it before then. This stage adds the child and, with it, the externalities, and then refits the model with the child in place.

**The child in the household.** Each household raises one child during its adulthood period. The child shares the household's consumption and housing through the equivalence scale $\xi$, while the drug only enters the utility of the parent, so the parent and the child hold different bundles,

$$X_P = \left[\left(\frac{c}{\xi}\right)^{1-\gamma}\left(\frac{h}{\xi}\right)^{\gamma}\right]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \left[\left(\frac{c}{\xi}\right)^{1-\gamma}\left(\frac{h}{\xi}\right)^{\gamma}\right]^{1-\theta} b^{\theta}, \tag{13}$$

where $X_P$ denotes the parent's bundle and $X_c$ the child's, and the parent's flow utility in period 1 is

$$u_1(c,h,v)\ =\ U(X_P)\ +\ \omega_c\,(\xi-1)\,U(X_c)\ +\ \aleph. \tag{14}$$

The child consumes the household's consumption and housing but none of the drug, so a parent's drug spending reaches it only by taking consumption and housing out of what they share, and it counts as $\xi - 1$ of the household against the scale. The weight $\omega_c$ is the share of the child's utility that the parent takes into account. Without the drug and with $\omega_c = 1$, the first two terms become $\xi\,U(X)$, which counts the household

[7] The share worse off is above the 8.95 percent who use under Housing First. Welfare is measured before a household learns its period 1 unemployment and voucher draws. Take a household with a given productivity that would use only after a certain combination of draws, for example only if it lost its job. Before the draws, it is worse off under Treatment First, because it may end up in that combination, but it counts among the users only in the cases in which it does. A household that would not use under any combination is not worse off.

[8] Spending what Treatment First saves on a higher $\pi_V$, so that the two programmes cost the same, raises $\pi_V$ by only 0.31 percent. That makes 40.4 percent of households that never use and may draw a voucher slightly better off, but those that would use under Housing First still lose, and on average households would still give up 0.0083 percent of consumption to keep Housing First, about half of the 0.017 percent at the same $\pi_V$ as in the benchmark of stage 1.

as $\xi$ adult equivalents: the parent as one and the child as the remaining $\xi - 1$. In period 2, when the parent moves into old age, the child leaves and the parent lives alone, meaning that $\xi = 1$, and flow utility is that of (2).

**The child leaves.** At the end of period 1 the child leaves the household and becomes an adult, entering period 1 of its own adulthood, with its own child living with it. Its productivity is correlated with its parent's, and it is further permanently reduced if the parent acquired the disorder while the child lived with it,

$$\log y_{\text{child}} \;=\; \rho \log y + (1-\rho)\mu_y + \varepsilon - \delta_d \, \mathbf{1}\{d' = 1\}, \qquad \varepsilon \sim \mathcal{N}\big(0,\, (1-\rho^2)\,\sigma_y^2\big), \tag{15}$$

where $\rho$ is the persistence of productivity across generations, $\mu_y = -\sigma_y^2/2$ is the mean of log productivity, which keeps its centre and the variance of $\varepsilon$ its spread the same in every generation, and $\delta_d$ is the earnings penalty for the parent's disorder.[9]

**The parent's problem.** The child enters the parent's problem twice, once for each part of its life. In childhood it is the second term of (14). As an adult it is the continuation value $D(y, d')$, which measures what the parent's disorder costs its child in the child's own lifetime value,

$$D(y, d') \;=\; \mathbb{E}\big[\bar{V}_1(y_{\text{child}}) \,\big|\, y,\, 0\big] - \mathbb{E}\big[\bar{V}_1(y_{\text{child}}) \,\big|\, y,\, d'\big] \;\geq\; 0, \tag{16}$$

where $\bar{V}_1$ is the value of entering period 1, before its draws, at the child's productivity from (15). The first term is for a parent without the disorder and the second for the parent's own $d'$. So $D(y, 0) = 0$, and the loss is $D(y, 1)$ when the parent's use leads to the disorder. $D(y, d')$ captures the direct impact of the parent's SUD on its child's utility, which runs through the child's earnings. It enters as a difference and not a level, so it costs the parent nothing unless its use makes the disorder possible. The parent's period-1 problem in (11) in Section 3.2.4 then becomes

$$V_1(y, e, \eta) = \max_{\{k,\, v \geq 0,\, a' \geq 0\}} \Big\{ u_1(c, h_k, v) + \beta \, \mathbb{E}_{d'} \Big[ \phi(d')\, V_2(a', y, d') \Big] - \;\beta \, \omega_c \, \mathbb{E}_{d'} \big[ D(y, d') \big] \Big\}. \tag{17}$$

$\omega_c$ multiplies both child terms. A planner who counted the child fully would put a weight of one on both, so what the parent leaves out, $1 - \omega_c$ of the child's terms, is the externality on the child. The expectation weights $D(y, 1)$ by $\pi(v)$, the chance that the use turns into a disorder. $V_1$ depends on $D$ through the parent's problem and $D$ on $V_1$ through the child's problem, so the two are solved as a fixed point.

**Calibrating the child.** With the child added, $\xi$, $\rho$, $\delta_d$ and $\omega_c$ need to be calibrated as well. I set $\xi$ to $\sqrt{2}$, the square-root scale for a household of two (OECD, 2013). The persistence of productivity across generations, $\rho$, is set to 0.50, within the range of the intergenerational elasticities of earnings in Section 2. The penalty $\delta_d$ is assigned the value 0.27 in logs to give the 24-percent earnings loss Nilsson (2017) measures for children exposed in utero to a temporary Swedish policy that let grocery stores in two regions sell strong beer, which made alcohol easier to buy during pregnancy. The parent's weight on its child $\omega_c$ is fitted so that switching the child's terms off, with every other parameter held, raises drug use by 18.3 percent, consistent with adults without children in NSDUH 2024 using 18.3 percent more than parents. Every other stage 1 parameter is refitted with the child in place, and Table 2 reports both.

**The optimal level of drug use.** I begin by examining what the optimal degree of drug use and SUD is. I do so by setting the weight a parent puts on its child to one, $\omega_c = 1$, so that a parent counts its child fully, with nothing else changed and the programme left as it is. In this case, no parent uses at all: participation and the SUD rate both fall to 0.00 percent, and homelessness to 0.55 percent.

[9] The penalty runs through income rather than through anything the child inherits about the drug. Letting the child inherit a greater susceptibility, a higher $\pi_d$, would be inconsistent with the data: a child facing a higher chance of converting use into a SUD optimally uses *less* ex ante, while children of parents with a SUD are *more* likely to use, not less. Income works on both counts, and it works through the channels explored in Section 3.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the empirical evidence reviewed in Section 2. The income gradient does the whole job, so the model transmits a SUD across generations without the child inheriting any greater susceptibility to the disorder.

**Treatment First with the child counted.** Counting the child's utility in welfare, as Section 6.1 describes, a newborn is better off in the long run under Treatment First than under Housing First, by 0.44 percent of lifetime consumption. Treatment First cuts participation from 9.66 to 7.16 percent and the share carrying the disorder from 4.00 to 2.96 percent, both at each policy's rest point, so fewer children enter adult life with the penalty. Breaking the gains by periods of life shows a rather even split: the gain is 0.20 percent in childhood, because parents spend more of their resources on them instead of on drugs, and 0.24 percent in adult life, because the child enters with a better productivity draw. Treatment First lowers homelessness too, by about 1 percent, because fewer households have a SUD and lose their jobs, and fewer are themselves poor for having been born to a parent with the disorder, so their income is higher.

## 3.8 The transition

Comparing rest points hides a key tradeoff. Normally, the debate sees Treatment First as taking immediately and giving back only later. In this model, this happens too: at implementation a household with a SUD can do nothing about its condition but loses its voucher, and it is then at a higher risk of homelessness, whereas lower substance use pays off only for the generations after it. However, in the stylized model, a reduction in drug use actually benefits children even upon impact, and no child is worse off while living at home: children live with their parents only before the parents can have a SUD, so a parent can still respond to the reform by stopping, and its child then escapes the earnings penalty and consumes more at home. In the full model, children do live with parents who have a SUD, and indeed some children are worse off and some become homeless, but most children benefit and on net the gain is large.

The reform moves the programme from Housing First to Treatment First at the same $\pi_V = 0.27$, in the model with the child, and every generation alive when it lands is counted, each over its own remaining life and against what the old rule would have given it. Figure 2 shows the paths for drug use, the disorder and homelessness of each generation as adults, from the adults at the reform on, against the Housing First rest point. It also shows the welfare of every generation: the old and the adults alive at the reform, their children, and the cohorts born after them.

The old, in their second period, chose under the old rule, carry whatever disorder that choice left them and cannot recover in a terminal period; conditionality now revokes the voucher from income-eligible households carrying a SUD that happened to draw one, and they lose 0.07 percent of consumption, with 0.74 percent of them homeless against 0.73 under the old rule. The adults, in their first period with their children at home, who were born into the old distribution, choose under the new rule; they lose 0.019 percent over their adult lives, which is the price of an option removed.

Their children, the first cohort born into the reform, gain 0.34 percent: 0.24 in adult life, since they are the first to enter it with the draws of parents who chose under Treatment First, and 0.094 in childhood, against the 0.20 that childhood contributes in the long run, since the parents they grow up with were themselves born into the penalised distribution. From the second cohort on the chain is at its new rest point and every cohort gains the long-run 0.44 percent. The cost falls on the adults and the old alive at the reform, and the benefit on the children born at the reform and, even more, on every cohort born after them.

## 3.9 Stage 3: the disamenity of homelessness

Before moving onto the full model, I take up the aggregate homelessness externality. The externality is modelled such that homelessness imposes a cost on each household's flow utility, reducing it by $\kappa_x H$, where $H$ is the share of households that is homeless, which every household takes as given, so the term moves no choice and the calibration of stage 2 stands unchanged. It enters welfare only, and the child bears it like everyone else, once for each period.

$\kappa_x$ is solved inside the model from the annual willingness to pay for ending homelessness in the contingent valuation of Loubière et al. (2020), 0.2 percent of income. Households in the model spend 73 percent of their income on consumption, so 0.2 percent of income translates to 0.27 percent of consumption. $\kappa_x$ is the disamenity per unit of homelessness, in utils per period alive, that costs an adult in the stage 2 economy as

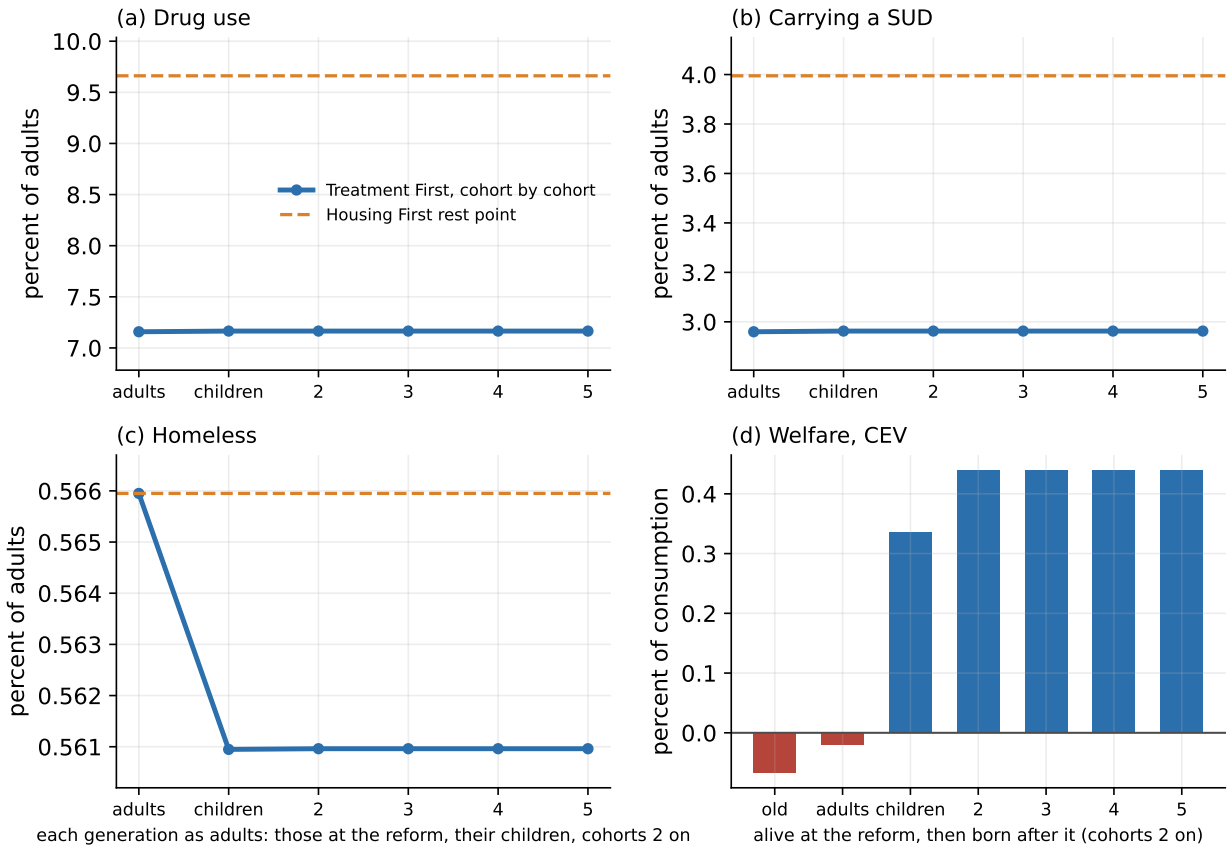



Figure 2: The transition from Housing First to Treatment First

Notes: the model with the child. Panels (a) to (c) are each generation as adults, in their first period: the adults at the reform, who were born into the Housing First distribution and choose under Treatment First, their children, and the cohorts born after them, when they become adults, to the fifth; the solid blue line is Treatment First, cohort by cohort, and the dashed orange line is the Housing First rest point. Panel (d) is each generation's CEV against Housing First: the old and the adults alive at the reform over the rest of their lives, and their children and every cohort born after them over their whole lives.

much over its adult life as giving up 0.27 percent of its consumption in every period.[10] That gives $\kappa_x = 1.30$. Because the two policies differ little in homelessness, it changes essentially nothing about the welfare gain from Treatment First in the stylized model, which remains at 0.44 percent of lifetime consumption.

# 4 The quantitative model

By now, I have shown how the main mechanisms in this paper work. This section describes the model the paper's numbers come from. It is very similar to the stylized model of Section 3, but includes annual periods and pieces that model leaves out: income risk during working age, retirement, progressive income taxes, interest on savings, a drug-use decision in every year, recovery, relapse and treatment, SMI, a higher risk of dying, worse employment prospects and a lower chance of recovering while homeless, children who can live with a parent who carries a SUD, a drug test that misses some users, a balanced government budget and an endogenous drug price. Notation follows Section 3 wherever the two models share an object: $\gamma$ and $\theta$ are the housing and drug weights in both, $\sigma$ the same risk aversion, $b$ is the same Stone-Geary shifter, $\aleph$ the same flow value of being alive, $\xi$ the same equivalence scale, $\omega_c$ the same weight on the child, $\rho$ and $\delta_d$ the same

[10] $\kappa_x$ solves $V\big((1-w)\,c;\,0\big) = V\big(c;\,H\big)$, where $V(c;H)$ is an adult's lifetime value in the stage 2 economy when it consumes $c$ and homelessness is $H$, which it bears as the disamenity $\kappa_x H$, and $w$ is the willingness to pay as a share of consumption to have $H = 0$.

persistence of productivity and penalty, $\psi_{ui}$ and $\bar{y}$ the same replacement rate and earnings cap, $\kappa_x$ the same disamenity, $\pi_d$ the same chance that use starts a SUD, $\kappa_{\phi d}$ the same mortality multiplier for the disorder, $h_1$ the same shelter, $\kappa_V$ and $\pi_V$ the same means test and chance of drawing a voucher, $p_r$ the same rent, and $\beta$ is the same discount factor. The values are in Section 5.

## 4.1 Demographics and the state and choice sets

Households enter the economy as adults at model age $j = 1$, live at most $J$ annual periods, work while $j < J_R$ and are retired from $J_R$. Each has a child, who lives in the household for the first $J_K$ years of its adult life and then leaves to become an adult with a child of its own. Each period a cohort of unit mass enters at $j = 1$, made up of the children who leave home that period, at their parent's death if that comes before $J_K$ and at $J_K$ otherwise (equation (38)), so the mass of households is constant in a stationary equilibrium. The individual state is age $j$ together with

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathbb{R}_+ \times \mathcal{Y}_j \times \{1,2,3\} \times \{0,1,2\} \times \{0,1\} \times \{0,1,2\}, \tag{18}$$

where $a$ denotes assets, $y$ productivity, $e$ employment status, $d$ the SUD, $m$ SMI and $\eta$ the household's standing with the voucher programme. The SUD takes three values because the parent's valuation of its child's prospects depends on its history and not only on its present (Section 4.11): $d = 0$ denotes a household that has never carried the disorder, $d = 1$ refers to one that carried it and has recovered, and $d = 2$ stands for one that carries it now. The voucher state is $\eta = 0$, denoting having no voucher, $\eta = 1$, representing holding one, and $\eta = 2$, standing for exclusion from the programme after failing Treatment First's condition.

Each period the household chooses the bundle

$$\big(h,\, v,\, \tau,\, a'\big) \in \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathbb{R}_+, \tag{19}$$

where $h$ is the dwelling, of which $h_1$ is a shelter, $v$ is drug consumption, $\tau$ is whether or not a household is in treatment for SUD, available only to a household that carries the disorder, and $a'$ is its next period's assets. Labour is inelastically supplied when employed.

## 4.2 Preferences

Flow utility is the stylized model's, with the child present for the first $J_K$ years. Consumption $c$ and housing $h$ are shared through the equivalence scale $\xi_j$, which scales for two members while the child is at home and for one after, and the drug enters the parent's utility alone, so the parent and the child hold the bundles

$$X_P = \Big[\Big(\frac{c}{\xi_j}\Big)^{1-\gamma}\Big(\frac{h}{\xi_j}\Big)^{\gamma}\Big]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \Big[\Big(\frac{c}{\xi_j}\Big)^{1-\gamma}\Big(\frac{h}{\xi_j}\Big)^{\gamma}\Big]^{1-\theta} b^{\theta}, \tag{20}$$

and period utility is

$$u\big(c,h,v,\tau\big) \;=\; U(X_P) \;+\; \omega_c\,(\xi_j - 1)\,U(X_c) \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u}, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{21}$$

$\gamma$ is housing's share of the non-drug bundle and $\theta$ the drug's share of the whole, $\sigma$ is the coefficient of relative risk aversion, $\aleph$ is a flow value of being alive, counted once per household, and $\bar{\tau}^{u}$ is the utility cost of a year in treatment, which is described in Section 4.6. The child's term is its own flow utility, which the parent weighs by $\omega_c$, and is counted as $\xi_j - 1$ of the household; it gets none of the drug, so a parent's drug spending reaches it through a reduction in the consumption and housing they share. After $J_K$ periods the child leaves to form its own household, and the child term is gone, at which point $\xi_j = 1$ for the parent.

Aggregate homelessness $H$ carries the disamenity $\kappa_x H$ for every household and child alive, which is what makes homelessness an externality for everyone rather than only a private misfortune. However, it does not affect any household's decision: each takes $H$ as given. Section 5.1 takes up its size.

## 4.3 Income, taxes and transfers

**Productivity.** Log productivity follows a persistent process over working life, $z_{j+1} = \rho_z z_j + \zeta_{j+1}$ with $\zeta \sim \mathcal{N}(0, \sigma_\zeta^2)$, and is held fixed at its last working value in retirement. Earnings on the job are $y\,\varepsilon(j)$, with $y = e^z$ normalised so that $y$ averages one at entry and $\varepsilon(j)$ the deterministic age profile. A household's productivity at entry is its inheritance from its parent, set out in Section 4.14; its spread combines $\sigma_\zeta$ with a persistent component drawn at entry, and a child entering with the penalty of that section is on a separate block of the grid, with the base grid scaled down for life.

**Before-tax income.** Before-tax income depends on age and employment,

$$Y(j,y,e) = \begin{cases} y\,\varepsilon(j) & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi_{ui}\ \min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(insured, first year out of work)} \\ 0 & j < J_R,\ e = 3 \quad \text{(out of work, uninsured)} \\ S(y) & j \geq J_R \quad \text{(retired)}. \end{cases} \tag{22}$$

UI replaces a share $\psi_{ui}$ of earnings on a base capped at $\bar{y}$, and it is paid only to a share $p_{ui}$ of those losing their job, mimicking the share of insured separations. After the first year, unemployed households, insured and uninsured alike, receive zero UI ($e = 3$). Social security $S(y)$ follows the statutory formula.[11]

The SUD and SMI do not cut earnings on the job; they act on whether there is a job at all, through the employment transitions below. Out of work, a household with the SUD or SMI is paid what any other household is: UI in its first year with probability $p_{ui}$, nothing with probability $1 - p_{ui}$, and nothing after the first year. There is no disability insurance in the model, so SMI brings no income of its own.[12]

**Taxes.** Before-tax income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate, so that after-tax income is

$$y^n = \min\left\{Y,\ \tau_\ell\, Y^{\varphi}\right\}\ +\ (1 - \tau_k)\, r\, a, \tag{23}$$

where $\tau_\ell$ scales after-tax income, and so sets the level of the tax, which falls as $\tau_\ell$ rises, and $1 - \varphi$ sets its progressivity; $\tau_k$ is the flat tax rate on capital income and $r$ is the exogenous interest rate on savings.[13]

**Transfers.** Food assistance pays the maximum allotment $\underline{c}$ less the share $\varrho$ of income, and the voucher pays a holder that is eligible this period the rent on the dwelling above a tenant contribution of $\kappa_r$ of income, up to the fair market rent (*FMR*),

$$\begin{aligned} F &= \max\left\{0,\ \underline{c} - \varrho\,\hat{y}\right\}, \\ Vch &= \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{\hat{y} < \kappa_V y^{\text{med}}\}\,\mathbf{1}\{h \neq h_1\}\ \min\Big\{\max\{0,\ p_r h - \kappa_r \hat{y}\},\ \max\{0,\ FMR - \kappa_r \hat{y}\}\Big\}, \end{aligned} \tag{24}$$

where $\hat{y} = Y + ra$ is the income both programmes are means-tested on, $\kappa_V$ and median income $y^{\text{med}}$ are the eligibility rule in Section 4.9, and *FMR* is the average rent paid by housed households in the benchmark. A voucher holder in a shelter keeps the voucher and is paid nothing until it rents again, as long as it keeps its eligibility. Voucher holders keep their vouchers for as long as they remain eligible. Each entering household receives a one-time endowment $T_0$ from bequests left by deceased households, and the government collects the rest of the estates of the dead.

[11] Let $\tilde{y} = \min\{y,\, y^{\max}\}$ denote the household's productivity at its last working age up to the taxable maximum $y^{\max}$, the ceiling above which earnings no longer raise the benefit. Then social security payments are $S(y) = \bar{w} \max\left\{\underline{S},\ \alpha_1 \min\{\tilde{y}, k_1\} + \alpha_2 \max\{0, \min\{\tilde{y}, k_2\} - k_1\} + \alpha_3 \max\{0, \tilde{y} - k_2\}\right\}$, where $k_1 < k_2$ are the two bend points, $\alpha_1 > \alpha_2 > \alpha_3$ the shares replaced below, between and above them, $\underline{S}$ the Supplemental Security Income (SSI) payment, and $\bar{w}$ the benchmark's average labour income. Their values are given in Section 5.2.

[12] In the data only a minority of adults with SMI receive disability benefits on the basis of a mental disorder: about 3 million, through Social Security Disability Insurance or SSI, against 14.6 million adults with SMI in NSDUH 2023, roughly one in five.

[13] The minimum keeps the labour income tax from turning negative, since at low incomes $\tau_\ell Y^{\varphi}$ exceeds $Y$ itself.

## 4.4 The budget constraint

The household's budget constraint is as follows:

$$c + p_v v + a' \;=\; y^n + a + T_0 \mathbf{1}\{j = 1\} \;-\; p_r\, h\, \mathbf{1}\{h \neq h_1\} \;+\; Vch + F, \qquad c \geq F,\; v \geq 0,\; a' \geq 0. \tag{25}$$

Food assistance is paid in kind, so it cannot directly fund the drug, the rent or saving. The drug price is $p_v$, which is endogenous here as opposed to exogenous like in the stylized model, and the rent per unit of housing $p_r$ is fixed and held at its benchmark value in every simulation. Treatment costs utility only.

## 4.5 Housing and the homelessness loop

Housing is the four-point grid of (19): $h_1$ is literal homelessness, sleeping rough or in an emergency shelter, which is free, and $h_2 < h_3 < h_4$ are three rental sizes.

Homelessness then enters three separate equations: survival (29) and the employment transitions (30), both in Section 4.8, and the SUD recovery hazard (27) in Section 4.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. SMI is the one thing it does not reach, for the reason given in Section 4.7.

## 4.6 Drug use, the SUD, recovery and treatment

A household that does not carry the disorder, either $d = 0$ (never had it) or $d = 1$ (had it but recovered), and uses acquires it ($d = 2$) with probability $\pi_d$,

$$\Pr(d' = 2 \mid d \neq 2,\, v) = \pi_d\, \mathbf{1}\{v > 0\}, \tag{26}$$

and a household that has recovered can acquire the disorder again. A household may begin using only up to an age $J_E$.[14] A household with the SUD recovers, to $d = 1$, with probability

$$\begin{aligned} \pi_r(h, m, \tau) = \min\Big\{1,\; \bar{\pi}_r \left(\mathbf{1}\{h = h_1\}\, \mu_H + \mathbf{1}\{h \neq h_1\}\right)\left(\mathbf{1}\{m = 1\}\, \mu_m + \mathbf{1}\{m = 0\}\right) \\ \times \left(\mathbf{1}\{\tau = 1\}\, g + \mathbf{1}\{\tau = 0\}\right)\Big\}, \end{aligned} \tag{27}$$

where $\bar{\pi}_r$ is the baseline annual chance of recovering while housed, without SMI and untreated. It has three multipliers on it: $\mu_H < 1$, which means that recovery is harder in a shelter, $\mu_m < 1$, which means that it is harder with SMI, and $g > 1$, which means that it is more likely in treatment.

$\mu_H$ is how housing acts on SUD recovery here. Housing does not change what a household wants: flow utility is the same function of $(c, h, v)$ housed or not. What housing changes is the chance of getting out of the disorder once it is carried. Milby et al. (2005) randomised cocaine-dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies $\mu_H$ in the calibration.

Treatment $\tau$ is a choice, taken at the utility cost $\bar{\tau}^u$ in (21), and $g$ is the only thing it buys. It does not stop the household using, it does not remove the excess mortality of the disorder while it lasts, and it does not exempt the household from Treatment First's condition.

## 4.7 SMI

$m$ is a two-state Markov chain representing SMI, mental illness severe enough to cause substantial functional impairment. It affects a small share of adults, whereas any mental illness affects roughly a quarter of adults and includes mild anxiety. Onset follows an age profile $\varsigma_m(j)$, normalised to one at age 26 and scaled by the

[14] Without it there is little-to-no consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it. The cause is that death is certain at the last age $J$, so close to it a household knows how little life remains, whereas in real life the vast majority of people do not know exactly when they will die. Section 5 gives the value of $J_E$.

level $\omega_m$, and a household that carries the SUD is $\iota > 1$ times more likely to develop SMI than one that does not,

$$\Pr(m' = 1 \mid m = 0, d) = \min\left\{1,\ \omega_m \varsigma_m(j)\left(\mathbf{1}\{d = 2\}\,\iota + \mathbf{1}\{d \neq 2\}\right)\right\}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m. \quad (28)$$

The effect of the SUD on SMI onset, $\iota > 1$, runs in the direction the evidence supports. About a quarter of those with a psychosis induced by a substance go on to develop schizophrenia or bipolar disorder (Murrie et al., 2020), and in the Danish registers people with any SUD are more likely to develop schizophrenia than people with no SUD, and those with an opioid use disorder more likely still (Nielsen et al., 2017).

SMI then affects three objects: the chance of finding a job when unemployed, through (30), mortality, through (29), both in Section 4.8, and the SUD recovery hazard, through $\mu_m$ in (27) in Section 4.6. It does not change the utility function or earnings on the job.

The SMI chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the moderate-needs arm of the At Home / Chez Soi randomised trial, which followed 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management had no detectable effect on symptom severity over twenty-four months (Stergiopoulos et al., 2015).

## 4.8 What the SUD, SMI, and homelessness cost

**Death risk.** Write $1 - \bar{\phi}_j$ for the probability of dying at age $j$ of a household that is housed and has neither the disorder nor SMI; the three multipliers below raise it for every other household. Write $q_j^0(h) = \min\{1, \left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}_j)\}$ and $q_j^1(h) = \min\{1, \kappa_{\phi d}\left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}_j)\}$ for the probability without and with the disorder. The probability of dying is then

$$1 - \phi(j, h, d, m) = \min\left\{1,\ \left(\mathbf{1}\{m = 1\}\,\kappa_{\phi m} + \mathbf{1}\{m = 0\}\right) q_j^0(h) + \mathbf{1}\{d = 2\}\left[q_j^1(h) - q_j^0(h)\right]\right\}, \quad (29)$$

so $\kappa_{\phi H} > 1$ multiplies the death probability when a household is in a shelter, $\kappa_{\phi d} > 1$ when a household has a SUD, and $\kappa_{\phi m} > 1$ the baseline risk when a household has SMI, to which the excess risk of the disorder is added. A household without the disorder therefore dies with probability $q_j^0(h)$, or $\min\{1, \kappa_{\phi m}\, q_j^0(h)\}$ with SMI, and the disorder adds the excess $q_j^1(h) - q_j^0(h)$ on top. The disorder kills whether or not it is being treated.[15]

**Unemployment risk.** A job is lost with probability $\pi^{eu}$ and found with probability $\pi^{ue}$,

$$\begin{aligned} \pi^{eu}(y, h) &= \min\left\{1,\ \bar{\pi}^{eu}\,\varpi(y)\left(1 + \mathbf{1}\{h = h_1\}\,\kappa_H\right)\right\}, \\ \pi^{ue}(h, d', m') &= \min\left\{1,\ \frac{\bar{\pi}^{ue}\left(1 - \mathbf{1}\{h = h_1\}\,\kappa_H\right)}{\left(\mathbf{1}\{d' = 2\}\,\kappa_d^{ue} + \mathbf{1}\{d' \neq 2\}\right)\left(\mathbf{1}\{m' = 1\}\,\kappa_m^{ue} + \mathbf{1}\{m' = 0\}\right)}\right\}, \end{aligned} \quad (30)$$

with $\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$ the baseline separation and job finding rates, $\varpi(y)$ a factor that makes separations likelier at low productivity, in line with the unemployment rates of the education groups the productivity grid spans, and the divisors $\kappa_d^{ue}$ and $\kappa_m^{ue}$ making one harder to find with the SUD and with SMI. Neither the SUD nor SMI raises separations; each only reduces the chance of regaining employment. $\kappa_H$ does both for a household in a shelter.

**The child's productivity penalty.** A child whose parent had the disorder at any point while the child lived at home, before $J_K$ or before the parent died if that came first, enters the economy with its productivity lowered by the proportion $\delta_d$ in (39) in Section 4.14, and the penalty stays with the child for life. The parent weighs the penalty through the weight $\omega_c$ it puts on its child (Section 4.11): using does not pass the penalty on by itself, but it carries the chance $\pi_d$ of acquiring the disorder, and with it the penalty. The penalty

[15] In the data, the meta-analyses of mortality among people using extramedical opioids report overdose as the largest single contributor to the excess (Larney et al., 2020; Degenhardt et al., 2011), and in the Centers for Disease Control and Prevention (CDC) WONDER database drug poisoning alone accounts for a large share of all deaths at ages 25 to 44, which is the moment Section 5.3 reports the model against.

is a risk to the parent's own objective only to the extent that $\omega_c$ is positive, so it carries an externality rather than a pure personal cost, and it is the one risk that reaches the child after it has left home; parental unemployment reaches the child only while it is at home, through the reduction in household resources, which the parent likewise weighs only at $\omega_c$.

## 4.9 The voucher and the two policies

A household is eligible for a voucher while its income is below the share $\kappa_V$ of median income,

$$\hat{y} \;<\; \kappa_V \, y^{\text{med}}, \tag{31}$$

with $\hat{y}$ the income of (24). A holder keeps its voucher while it is eligible, so unlike Section 3.2.3 the voucher is persistent rather than redrawn each period, and an eligible household without one draws one with probability $\pi_V$. Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses or carries the disorder, $v > 0$ or $d = 2$, whether or not it is in treatment. Income-eligible holders and income-eligible households that draw a voucher are tested, at a cost $c_{\text{test}}$ each to the government, and one that fails is detected with probability $p_{\text{det}}$, so unlike in Section 3.5, enforcement is not perfect: with probability $1 - p_{\text{det}} > 0$ a household that fails the condition actually gets to keep its voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. Either way the household is excluded from the voucher programme for a period of time, $\eta' = 2$: it cannot draw a voucher whatever its income, and it returns to the lottery, $\eta' = 0$, with probability $\psi_x$ each year.

None of $p_{\text{det}}$, $\psi_x$ and $c_{\text{test}}$ binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

## 4.10 The shocks, collected

Conditional on the state and the choices, the next period's state is drawn in five transitions, which are independent given the state except that employment depends on next period's SUD and SMI and on the dwelling $h$ through (30) of Section 4.8:

**Productivity $y$.** By the process of Section 4.3 while working, and unchanged in retirement.

**Employment $e$.** A household that loses its job is insured with probability $p_{ui}$, for one year. With $\pi^{ue}(h, d', m')$ and $\pi^{eu}(y, h)$ from (30),

$$e = 1 \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu}(y,h) \\ e' = 2 & \text{w.p. } p_{ui}\,\pi^{eu}(y,h) \\ e' = 3 & \text{w.p. } (1 - p_{ui})\,\pi^{eu}(y,h) \end{cases} \qquad e \in \{2,3\} \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } \pi^{ue}(h,d',m') \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}(h,d',m'). \end{cases} \tag{32}$$

State 2 is reachable only from employment and never from itself, so the insured are in it for the first year out of work and then move to state 3, which pays nothing.

**The SUD $d$.** A household without the disorder acquires it with the probability in (26) and one with it recovers with the probability in (27), both in Section 4.6, conditional on the chosen $v$, $h$ and $\tau$ and on $m$.

**SMI $m$.** SMI begins and ends according to (28) in Section 4.7, conditional on $d$ and age.

**The voucher $\eta$.** A voucher is drawn and revoked according to the rule of Section 4.9, conditional on this period's eligibility and, under Treatment First, on the chosen $v$ and the state $d$.

Write $\Gamma\big(s' \mid s, h, v, \tau, a'\big)$ for their joint law of motion, which is the object the expectation in (33) in Section 4.11 is taken against.

## 4.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot) = 0$,

$$
\begin{aligned}
V(j,s) = \max_{\{h,\,v,\,\tau,\,a'\}} \Bigg\{ & U(X_P) + \omega_c(\xi_j - 1)\,U(X_c) + \aleph - \tau\bar{\tau}^u \\
& + \;\beta\,\phi(j,h,d,m)\int_{\mathcal{S}} V(j{+}1, s')\,\Gamma\big(\mathrm{d}s' \mid s,h,v,\tau,a'\big) \\
& - \;\beta\,\chi_j\,\omega_c\,D(y,d)\Bigg\}, \qquad \chi_j = \begin{cases} 1-\phi(j,h,d,m) & j < J_K \\ 1 & j = J_K \\ 0 & j > J_K, \end{cases}
\end{aligned}
\tag{33}
$$

subject to the budget (25), the income and transfer rules (22) to (24) in Section 4.3, and the restrictions that treatment is available only to a household with $d = 2$, that no household begins using after $J_E$, and that $c \geq F$, since the food stamp payment is paid in kind and cannot be saved.

The first line is the flow of (21).[16] The second is the household's own continuation. The third is the child as an adult. It enters when the child leaves: at $J_K$ with certainty, and before then with the probability that the parent dies first. $D(y,d)$ is what the parent's disorder costs the child, in the child's own lifetime value,

$$
D(y,d) = \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ 0\big] - \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d\big] \;\geq\; 0, \tag{34}
$$

where the child is valued entering at $j = 1$ with the endowment $T_0$, employed, and without the disorder, SMI or a voucher. Its productivity is drawn as in (39) in Section 4.14 from a parent with productivity $y$. In the first term the parent has never had the disorder, and in the second it has its actual $d$, so $D(y,0) = 0$. They are charged ($D(y,d) > 0$ for $d \geq 1$) whenever they carried the disorder at some point before the child becomes an adult.

The term is a difference and not a level, and the distinction is not cosmetic. Written as a level, $\omega_c$ times the child's value, it would be worth more to a household the likelier it is to die, because it is the death branch that carries it, and a SUD raises the death risk severalfold, so a level would pay most to exactly the households the paper is about, for reasons that have nothing to do with the child. Instead, when it enters as a difference, the term is zero for a parent that has never carried the disorder and a cost otherwise, so what enters the decision is only the loss the parent's own history imposes on its child.

One weight $\omega_c$ multiplies both child terms, the flow of childhood and the value of adulthood, so calibrating it pins the parent's concern for the whole of the child's life at once. At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ it internalises the share $\omega_c$ and only $1 - \omega_c$ remains external. The policy functions $h(j,s)$, $v(j,s)$, $\tau(j,s)$ and $a'(j,s)$ are the dwelling, drug use, treatment and saving that solve (33) at age $j$ and state $s$.

## 4.12 The government

The government pays social security, UI, food assistance and vouchers under Housing First, with the addition of the drug tests under Treatment First. It collects the labour and capital income taxes of (23) in Section 4.3 and the estates of the dead, less the endowment $T_0$ given to entering adults ($j = 1$),

$$
\underbrace{\textstyle\sum \big(Y - \min\{Y, \tau_\ell Y^{\varphi}\}\big) + \tau_k \sum ra + \big(\text{estates} - T_0\big)}_{\text{revenue}} \;-\; \underbrace{\big(S + UI + F + Vch + \text{tests}\big)}_{\text{outlay}} \;=\; \mathcal{R}. \tag{35}
$$

In the benchmark the tax parameters are set as in Section 5 and $\mathcal{R}$ is whatever surplus they produce. In every experiment the taxes are held at their benchmark values and the probability of drawing a voucher $\pi_V$ adjusts so that $\mathcal{R}$ equals its benchmark value $\bar{\mathcal{R}}$.

[16] The disamenity $\kappa_x H$ is not in the formulation, because no choice of the household changes it (Section 4.2).

## 4.13 The drug market

The price of the drug clears its market,

$$p_v = \nu_1 S_v^{1/\varepsilon_s}, \qquad S_v = \sum_{j=1}^{J} \int v(j,s)\, \mathrm{d}\Phi(j,s), \tag{36}$$

where $\varepsilon_s$ is the price elasticity of supply and $\nu_1$ is normalised so that the price is one in the benchmark, and then held fixed across experiments. The price itself is not fixed: it adjusts in every experiment to clear the drug market.

## 4.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi(j{+}1, \mathcal{S}') \;=\; \int_{\mathcal{S}} \phi\big(j, h(j,s), d, m\big)\, \Gamma\big(\mathcal{S}' \mid s,\, h(j,s), v(j,s), \tau(j,s), a'(j,s)\big)\, \mathrm{d}\Phi(j,s), \tag{37}$$

for $j = 1, \ldots, J-1$ and every measurable set $\mathcal{S}'$ of the state space $\mathcal{S}$ of (18), with the integral over the state $s$ of the households of age $j$, together with the inflow of entrants

$$\Phi(1, \mathcal{S}') \;=\; \frac{\sum_{j \le J_K} \int_{\mathcal{S}} \chi_j\, \mathcal{N}\big(\mathcal{S}' \mid s\big)\, \mathrm{d}\Phi(j,s)}{\sum_{j \le J_K} \int_{\mathcal{S}} \chi_j\, \mathrm{d}\Phi(j,s)}, \tag{38}$$

where the integrals are again over $s$, $\mathcal{N}$ places the child according to (39) and $\chi_j$ is the weight defined in Section 4.11. The entering cohort has unit mass every period, so in equilibrium it equals the mass of the dead. Where an entrant starts depends on its parent: its productivity is drawn around the parent's, and it carries the penalty if the parent had the disorder, both as of the moment the child becomes an adult, which is at the parent's death if that comes before $J_K$ and at $J_K$ otherwise, the same timing with which the parent weighs its child in Section 4.11.

An entrant starts with the endowment $T_0$, is employed, has never carried the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26. Its log productivity is drawn around its parent's, with the persistence $\rho$ and an innovation that keeps the spread of productivity at entry the same in every generation, and it is scaled for life by the penalty if the parent had the disorder at any point before its child became an adult,

$$\begin{aligned} z_1^{\text{child}} \;&= \rho\, z^{\text{parent}} + (1-\rho)\,\mu + \epsilon \\ &\quad + \log(1-\delta_d)\, \mathbf{1}\{\text{parent has } d \ge 1 \text{ when its child becomes an adult}\}, \end{aligned} \tag{39}$$

with $z_1^{\text{child}}$ the child's log productivity at entry, age $j = 1$, $z^{\text{parent}}$ the parent's log productivity when its child becomes an adult, $\mu$ the mean of log productivity at entry and $\epsilon$ normally distributed with mean zero and variance $1 - \rho^2$ times the variance of log productivity at entry, which keeps that variance the same in every generation. The draw is of log productivity, and the child's productivity is $y = e^{z_1^{\text{child}}}$, as in Section 4.3. Because the penalised child's own productivity is lower, its own child inherits a lower draw too, so the penalty also reaches the grandchild through the persistence.

## 4.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{h_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and

- prices, policies and aggregates $\{p_{vt},\, \pi_{Vt},\, H_t\}_{t=0}^{\infty}$, together with $\bar{y}$, $y^{\text{med}}$ and $FMR$, which carry no date because they are kept fixed at their benchmark values,

such that, at every date $t$,

1. the policy functions solve (33) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the entrants of (38) and (39);
3. in every experiment, the government budget (35) holds at $\pi_{Vt}$ with $\mathcal{R} = \bar{\mathcal{R}}$;
4. the drug market clears at $p_{vt}$, so (36) holds;
5. the homelessness rate $H_t$ is the aggregate implied by $\Phi_t$.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the long-run policy comparisons report. The sequence version is what the transition between them solves.

# 5 Calibrating the quantitative model

The model has fifty-nine parameters, some of which vary in value by age. Thirty-seven are calibrated externally, from data or from the literature, and twenty-two are calibrated internally. Three further parameters govern drug testing and exclusion from the voucher; they do not play a role in the benchmark, where nothing is conditional, but they do play a role under Treatment First, which is where they are set. The calibrated parameters are split across the two tables of this section: twenty-six are key to the main story of this paper and its policy analysis (Table 4, sixteen parameters internally and ten externally calibrated), and the other thirty-three are standard (Table 5, six parameters internally and twenty-seven externally calibrated).

The tables carry the parameter, its description, the moment and source used to fix its value, the value of that moment in the data and in the model, and the value of the parameter itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and how its moment identifies it. `data/DATA_APPENDIX.md` in the replication package documents every estimate in full, naming the dataset behind each target, its year and universe, and the computation that produces it. The package carries the extracts themselves alongside the code that reads them, so nothing has to be downloaded.

## 5.1 The twenty-six parameters that carry the story of this paper

Of the fifty-nine parameters, twenty-six are specific to this paper and carry its main story and policy analysis; the other thirty-three, common to most life-cycle models, are in Section 5.2. The former are set out in Table 4, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table. For clarity the table groups them into numbered blocks, each collecting the parameters that work through one mechanism, so that what each parameter does and how it moves the model can be read alongside the others that act along similar lines; the blocks are taken in turn below, and those of the standard parameters in Section 5.2 continue the numbering.

**Block 1: Who uses drugs.** The five parameters in this block set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

**$\theta$, the weight on drugs in utility.** It is the parameter that determines the amount spent on drugs when the household consumes a strictly positive amount of it. As it rises use rises on both the extensive and intensive margins, which means that the externality rises and there is more for either policy to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent in the data, from retail drug spending in Midgette et al. (2019) and the incomes of adults aged 26 and over who use in NSDUH 2024.

**$b$, whether a household uses at all.** The marginal value of the drug goes as $\theta/(b{+}v)$, so a larger $b$ makes the first units worth less and a household with little to spend buys nothing. As $b$ declines use rises,

Table 4: The twenty-six parameters that carry the story of this paper

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Block 1: Who uses drugs* | | | | | | |
| $\theta$ | Weight on drugs in utility | Int. | Drug spending over income, Midgette et al. (2019) | 16.0% | 14.8% | 0.165 |
| $b$ | Whether a household uses at all | Int. | Any drug use disorder, NSDUH 2024 | 8.90% | 8.96% | 0.0054 |
| $\pi_d$ | Chance that a year of use leads to a SUD | Int. | Share of users carrying a SUD, NSDUH 2024 | 41.3% | 42.6% | 0.0345 |
| $\kappa_{\phi d}$ | Increased mortality with a SUD | Int. | Income of drug users over average, NSDUH 2024 | 74% | 62% | 4.00 |
| $\aleph$ | Flow value of being alive | Int. | VSL, Viscusi and Aldy (2003) | 80 | 81 | 87.6 |
| *Block 2: Costs of homelessness and vouchers* | | | | | | |
| $\kappa_d^{ue}$ | How much harder it is to regain employment with a SUD | Int. | SUD share of the homeless, Gutwinski et al. (2021) | 21.7% | 20.2% | 3.78 |
| $h_1$ | Service flow of homelessness | Int. | Homelessness rate, HUD PiT 2023, within a band | 0.40–1.00% | 0.62% | 0.0216 |
| $\kappa_{\phi H}$ | Mortality when homeless | Ext. | Meyer et al. (2023) | 1.60 | n.a. | 1.60 |
| $\kappa_H$ | Employment penalty of homelessness | Ext. | Desmond and Gershenson (2016) | 0.20 | n.a. | 0.20 |
| $\underline{c}$ | Food stamp maximum allotment | Int. | SNAP outlays over output | 0.42% | 0.43% | 0.0879 |
| $\pi_V$ | Chance of drawing a voucher | Int. | Recipients over eligible, HUD 2023 | 26.6% | 27.5% | 0.0675 |
| $T_0$ | Endowment at entry to adulthood ($j = 1$) | Int. | Share of homeless aged 25 to 34, HUD PiT 2024 | 26.0% | 26.4% | 0.111 |
| *Block 3: SMI chain* | | | | | | |
| $\iota$ | SUD multiplier on SMI onset | Ext. | Danish registers, Nielsen et al. (2017) | 2.00 | n.a. | 2.00 |
| $\lambda_m$ | Annual chance of recovering from SMI | Ext. | SMI ten-year recovery rate, Åsbø et al. (2022) | 50% | n.a. | 0.066 |
| $\omega_m$ | Level of the SMI onset profile | Int. | SMI share of the population, NSDUH 2024 | 4.82% | 4.83% | 0.0037 |
| $\kappa_m^{ue}$ | How much harder it is to regain employment with SMI | Int. | SMI share of the homeless, Gutwinski et al. (2021) | 25% | 24.7% | 7.71 |
| $\mu_m$ | SUD recovery multiplier for SMI | Ext. | Dual diagnosis relapse, Bradizza et al. (2006) | 0.82 | n.a. | 0.82 |
| $\kappa_{\phi m}$ | Mortality multiplier for SMI | Ext. | Olfson et al. (2015) | 2.50 | n.a. | 2.50 |
| *Block 4: Recovery and treatment* | | | | | | |
| $\bar{\pi}_r$ | Baseline SUD recovery rate | Ext. | SUD ten-year remission rate, Vaillant (1995) | 22% | n.a. | 0.025 |
| $\mu_H$ | SUD recovery when homeless | Int. | Abstinence, housed/homeless, Milby et al. (2005) | 1.47 | 1.45 | 0.696 |
| $g$ | Treatment effect on recovery | Ext. | Remission with medication for opioid use disorder | 2.00 | n.a. | 2.00 |
| $\bar{\tau}^u$ | Utility cost of treatment | Int. | Treatment take-up, NSDUH 2024 | 24.5% | 24.9% | 4.49 |
| *Block 5: Externalities* | | | | | | |
| $\omega_c$ | Parent's weight on its child | Int. | Non-parents' use over parents', less one, NSDUH 2024 | 18.3% | 17.7% | 0.085 |
| $\delta_d$ | Earnings the child loses to a parent's SUD | Ext. | Earnings lost in utero, Nilsson (2017) | 0.24 | n.a. | 0.24 |
| $\kappa_x$ | Disamenity of homelessness | Int. | Willingness to pay, Loubière et al. (2020) | 0.20% | 0.20% | 0.93 |
| *Block 6: Drug market* | | | | | | |
| $\varepsilon_s$ | Price elasticity of drug supply | Ext. | Mace et al. (2020) | 1.46 | n.a. | 1.46 |

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

raising the externality that policy can affect. Its value is chosen to target a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over, and the model gives 8.96 percent.

**$\pi_d$, the chance that a year of use leads to a SUD.** This is what makes using *risky*, and it is the step that turns drug consumption into a disorder. Raise it and the externality rises, and the moral hazard costs more, since it is $\pi_d$ that converts an extra voucher under Housing First into a higher SUD rate that Treatment First then attempts to prevent. Its value is chosen so the model generates the share of users who carry a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024.

**$\kappa_{\phi d}$, increased mortality with a SUD.** As discussed at length in the stylized model, this is a key parameter in the paper. Mortality truncates the value of life for a user, and as it rises, it hits households with the highest life values hardest and thus reverses the income gradient, making the poor the users rather than the rich, in equilibrium when $b$ is chosen to match the drug use rate. It is calibrated to the average income of users, setting it to 4.00, within the range that studies of excess mortality among drug users report: from hazard ratios of 1.27 for cocaine and 2.40 for heroin among those who have ever used them (Walker et al., 2017) to a standardised mortality ratio of 14.7 among people dependent on opioids (Degenhardt et al., 2011). At that value users earn 62 percent of average income in the model, against 74 percent in NSDUH 2024.

**$\aleph$, the flow value of being alive.** $\aleph$ is the other half of the mortality mechanism: $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying and $\aleph$ sets what the life being lost is worth. Increase it and use falls, because dying is more costly. Its value is chosen so the model generates a VSL of 80 times average labour income, which is the \$9m of Viscusi and Aldy (2003) against a mean household income of \$113,000.[17]

**Block 2: Costs of homelessness and vouchers.** Both Housing First and Treatment First work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to

[17] No household has a negative value of life in equilibrium, so death is never an optimal outcome.

a household in that state and what being removed from that state is worth.

$\kappa_d^{ue}$**, how much harder it is to regain employment with a SUD.** A household with a SUD who is unemployed finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_d^{ue}$. Raising $\kappa_d^{ue}$ makes unemployment spells longer for households with a SUD, and thus they are more likely to be unhoused, which is the link from the disorder to the street, and which also affects the efficacy of Treatment First to deter drug use. It is internally calibrated to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021), which the model gives as 20.2.

$h_1$**, the service flow of homelessness.** This parameter sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is smaller. When $h_1$ rises, a shelter is less bad, a voucher is worth less and Treatment First has less to act on. Its value is chosen so that the homelessness rate lies within 0.40 to 1.00 percent of households. The floor is HUD's 2023 PiT count, which counts a single night and so misses households homeless at other times of the year; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). I set $h_1$ internally so that the model delivers a number inside this range, 0.62 percent.

$\kappa_{\phi H}$**, mortality when homeless.** This parameter determines how much more likely a homeless household is to die because of its homelessness. Increasing it makes homelessness worse, raises the value of a voucher and makes Treatment First's deterrence larger. I set it externally so that it multiplies the baseline mortality by 1.60 at every age, the mortality ratio for homeless adults against poor housed adults in Meyer et al. (2023).

$\kappa_H$**, the employment penalty of homelessness.** This parameter decides how much more likely a household in a shelter is to lose its job and how much less likely it is to find one: separations are $1+\kappa_H$ times as frequent and job finding $1-\kappa_H$ times as fast. Housing First mechanically recovers employment while Treatment First only does so if, over time, the decline in SUD is large enough that fewer households are unemployed and homeless. As this parameter's value rises the cost of homelessness does too, and there is more for both Housing First and Treatment First to act on. It is set externally at 0.20, from Desmond and Gershenson (2016).

$\underline{c}$**, the food stamp maximum allotment.** This parameter determines how low consumption can go, which determines where utility is bounded from below for the poorest households with no income, no assets, and no vouchers. Raise it and the value of being alive rises, which increases the cost of homelessness and SUD, and gives more for Treatment First to deter. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP in the data, and over output, which in this model is labour income.

$\pi_V$**, the chance of drawing a voucher.** This is the policy dial itself. On the one hand, increasing it reduces the childhood and aggregate homelessness externalities, since more households are housed. On the other hand, increasing it increases the value of being alive, since life is worth more when housed, and thus it allows Treatment First to be more effective in deterring drug use and fixing the budget and earnings externalities: the whole trade in one parameter. In the benchmark it is calibrated internally to 0.068, so that 27.5 percent of the income-eligible population holds a voucher, against 26.6 percent in HUD's PSH, 2023.

$T_0$**, the endowment at entry to adulthood (**$j = 1$**).** A child who becomes an adult enters the economy with an endowment of $T_0$ times average earnings, which sets how likely a young adult is to end up in a shelter in its first years. Raise it and fewer young adults become homeless, and Treatment First would affect behaviour less during the child-rearing years, when parents' choices affect their children's outcomes. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count, which the model gives as 26.4.

**Block 3: SMI chain.** This block determines how much further Treatment First reaches over and above the disorder it prevents: a SUD raises the chance of SMI, and SMI lowers employment and raises death.

$\iota$**, the SUD multiplier on SMI onset.** The hinge of this whole block. Without it SMI is exogenous to everything a policy does and treating SUD cannot touch it. As it rises a SUD causes more SMI, and

Treatment First's return rises, because curing the disorder now prevents some cases of SMI too. It is set externally at 2.00, between the hazard ratios of 1.34 for any SUD and 2.74 for opioid use disorder in the Danish registers of Nielsen et al. (2017).

$\lambda_m$**, the annual chance of recovering from SMI.** This parameter governs how long an episode of SMI lasts: it is the hazard of leaving the SMI state each year. Raise it and SMI is less persistent, and Treatment First has less of a deterrent effect, since SMI spells, some of them caused by a SUD, are shorter on average. It is set externally at 0.066, the annual hazard that gives a 50 percent chance of recovery within ten years, the ten-year clinical recovery rate for bipolar disorder in the first-episode psychosis cohort of Åsbø et al. (2022).

$\omega_m$**, the level of the SMI onset profile.** This parameter scales the chance of developing SMI at every age, so it sets how many households have SMI. Increase it and the impact of a SUD on acquiring SMI is larger, and Treatment First has more to act on. It is calibrated internally, to the share of the population carrying SMI, 4.82 percent, which is the NSDUH 2024 prevalence by age weighted by the model's own age distribution, and the benchmark gives 4.83.

$\kappa_m^{ue}$**, how much harder it is to regain employment with SMI.** The same as $\kappa_d^{ue}$ but for SMI: a household with SMI finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_m^{ue}$. Increase it and the average unemployment spells during SMI periods are longer, and the utility cost of SMI is higher. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), which the model gives as 24.7, for the same reason as the SUD's: a household out of work for longer is more likely to eventually become homeless.

$\mu_m$**, the SUD recovery multiplier for SMI.** This parameter is the second direction of the chain: not only does a SUD increase the chance of SMI, but SMI also makes SUD more persistent. As it falls the disorder is harder to recover from, and Treatment First has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from the dual-diagnosis relapse rates in Bradizza et al. (2006), whose review covers people carrying both a SUD and a mood, anxiety, schizophrenia-spectrum or personality disorder.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and the one through which it shortens life. As it rises deaths among households with SMI rise, and Treatment First has more to work on, because reducing drug use reduces SUD and, through it, the chance of SMI and ultimately the chance of death. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

**Block 4: Recovery and treatment.** Both policies work through parameters in this block, since recovery is what Treatment First encourages directly and what Housing First buys by housing people, because recovery is more likely while housed. The first two set how much recovery there is to buy and the last two how much of it a policy can reach.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** This parameter is the hazard of recovering from a SUD for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner on their own, and there is less for either policy to buy. It is set externally from the ten-year remission rate in Vaillant (1995), which gives an annual hazard of 0.025.

$\mu_H$**, SUD recovery when homeless.** The value of this parameter determines how much less likely recovery from the disorder is for an unhoused household relative to a housed one. It determines how much a voucher buys beyond shelter. As it falls (i.e., SUD recovery while unhoused is lower) the gradient steepens, so housing buys more recovery and Housing First is more effective at curing SUD. Near one, Housing First has little effect and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine-dependent homeless adults: an average of 6.60 weeks abstinent in the arm housed without conditions against 4.48 in the arm given no housing, so housing raises recovery by 47.3 percent.

$g$**, the treatment effect on recovery.** This is the instrument through which Treatment First reaches households that already carry a SUD, which is what matters for short-run impacts. Raise it and treatment is more likely to cure a SUD and Treatment First has a larger margin it can affect. It is set externally to 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it, which is also what randomised trials of agonist medication against placebo

find for abstinence.

$\bar{\tau}^u$**, the utility cost of treatment.** Take-up is a choice, not a rule, and this is what makes households decline treatment they might gain from. When it rises, fewer opt for treatment, and Treatment First induces fewer households with a SUD to seek treatment. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024, and 24.9 percent do in the model.

**Block 5: Externalities.** The parameters in this block decide whether something is an externality and, if so, what its magnitude is. This affects the ranking between Housing First and Treatment First, since the no-externality comparison already favours Housing First, because households are rational and forward looking. The first two are how much of the child a parent counts and what a parent's disorder costs the child, and the third is what the homelessness of strangers costs everybody. All three concern externalities in the strict sense: one household's choice enters another household's utility directly without being internalised by the household making the decision.

$\omega_c$**, the parent's weight on its child.** A parent bears $\omega_c$ of what its choices cost its child, during their childhood and in the draw the child enters with. As it rises the parent counts more of the child's loss, so the budget, earnings and childhood homelessness externalities left for Treatment First and Housing First to correct are smaller. It is identified by a counterfactual moment: with every other parameter unchanged, drug use with the child given no weight is compared with drug use at $\omega_c$. In NSDUH 2024 adults aged 26 and over without children use at a rate 18.3 percent above parents', and at $\omega_c = 0.085$ removing the child's weight raises use by about the same in the model.

$\delta_d$**, earnings the child loses to a parent's SUD.** The child does not inherit the disorder, only a worse productivity draw, and that lower expected draw is what makes it likelier to be poor, to use, to acquire a disorder and SMI consequently, to be unhoused and to die earlier. When $\omega_c < 1$, raising it increases the earnings externality, and loads whichever policy prevents more drug use: Treatment First by deterring the parent from using, Housing First by making recovery more likely once housed. It is set externally at 0.24, the earnings loss Nilsson (2017) measures for children exposed in utero to a Swedish alcohol experiment, and with it the model delivers an odds ratio of 1.46 of acquiring a SUD for the child of a parent with one, relative to a child born to a parent who did not have one, within the range of 1.44 to 2.51 in the literature (Section 5.3).[18]

$\kappa_x$**, the disamenity of homelessness.** This parameter connects one household's housing status to the utility of another who is not its own child. Each household's flow utility falls by $\kappa_x$ times the aggregate homelessness rate, which no household can move on its own, so it moves no policy function and prices only what the policies do to the headcount. As it rises the aggregate homelessness externality increases, which loads whichever policy houses more people. The estimate is the willingness to pay for ending homelessness in the contingent valuation of Loubière et al. (2020), 0.20 percent of income. Its value solves the same problem as in the stylized model (Section 3.9), with that willingness to pay converted to a share of consumption.

**Block 6: Drug market.** Only one parameter of the drug market belongs here. It is what makes the drug price endogenous, so that when drug demand changes, so does the equilibrium drug price. This is a general-equilibrium offset that runs against Treatment First in particular. Deterring households from using takes them out of the drug market, demand falls and so does the price, and a cheaper drug draws other households in. Some of what the policy removes is handed back through the price, and how much depends entirely on this parameter.

$\varepsilon_s$**, the price elasticity of drug supply.** When it rises, supply is more elastic, so the price moves less when demand does, and the general-equilibrium offset to Treatment First shrinks: at a very elastic supply, curing households barely lowers the price, so general-equilibrium effects play little role. Thus, setting it to a low value loads the model against Treatment First. I set it to 1.46, which is what Mace

[18] The penalty carries only 11 percent of the welfare gain from Treatment First (Section 6.2), so the ranking of the two policies does not rest on its size.

et al. (2020) estimate for cannabis on Washington State administrative data,[19] which is at the lower end of elasticities used in the drug supply literature (Rhodes et al., 2002).

## 5.2 Standard parameters

The remaining thirty-three parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Six of them I calibrate internally, against a moment, because no source reads them off directly; the other twenty-seven I set from outside, from an estimate or from statute. Table 5 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the six I calibrate internally, since the other twenty-seven have no target to reproduce.

These parameters are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of productivity determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters above is not that they do nothing, but that they are not specific to this paper.

Table 5: Standard parameters

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Block 7: Preferences* | | | | | | |
| $\sigma$ | Risk aversion | Ext. | Attanasio and Weber (2010) | 2.00 | n.a. | 2.00 |
| $\gamma$ | Weight on housing services | Int. | Housing over expenditure net of drugs, CEX 2024 | 26.8% | 26.2% | 0.536 |
| $\beta$ | Discount factor | Int. | Wealth over output, BEA fixed assets | 3.00 | 2.95 | 0.946 |
| $\xi_j$ | Equivalence scale | Ext. | Square-root scale, OECD | 1.41 | n.a. | 1.41 |
| *Block 8: Rental ladder* | | | | | | |
| $h_2$ | Smallest rental size | Int. | Renters paying over half of income, SCF 2022 | 19.2% | 25.8% | 0.0407 |
| $h_3/h_2$ | Middle rung over the smallest | Ext. | Set | 4.00 | n.a. | 4.00 |
| $h_4/h_2$ | Largest rung over the smallest | Ext. | Set | 8.00 | n.a. | 8.00 |
| *Block 9: Earnings and employment* | | | | | | |
| $J$ | Maximum age | Ext. | Age 85 | 60 | n.a. | 60 |
| $J_K$ | Years the child lives at home | Ext. | Parent aged 26 to 50 | 25 | n.a. | 25 |
| $J_R$ | Retirement age | Ext. | Age 66, SSA full retirement age | 41 | n.a. | 41 |
| $\varepsilon(j)$ | Age profile of efficiency | Ext. | Hump peaking at age 56, Greenwood et al. (2025) | Profile | n.a. | Profile |
| $\rho_z$ | Persistence of productivity | Ext. | Guvenen et al. (2023) | 0.90 | n.a. | 0.90 |
| $\sigma_\zeta$ | Standard deviation of its shocks | Int. | Bottom half's share of earnings, SCF 2022 | 17.2% | 16.2% | 0.413 |
| $\rho$ | Persistence of productivity across generations | Ext. | Solon (1992), Mazumder (2005) | 0.50 | n.a. | 0.50 |
| $\bar{\pi}^{eu}$ | Job separation rate | Int. | Unemployment rate, BLS CPS 2014–24 | 4.9% | 5.6% | 0.0268 |
| $\varpi(y)$ | Separations by productivity | Ext. | Unemployment by education, BLS 2019 | Profile | n.a. | Profile |
| $\bar{\pi}^{ue}$ | Annual job finding rate | Ext. | Krueger et al. (2014) | 0.55 | n.a. | 0.55 |
| $p_{ui}$ | Share of separations that are insured | Ext. | UI applicants among the unemployed, BLS 2022 | 0.30 | n.a. | 0.30 |
| *Block 10: Baseline survival and SMI onset* | | | | | | |
| $\bar{\phi}_j$ | Baseline survival by age | Ext. | US life table 2010, Arias (2014) | Profile | n.a. | Profile |
| $\varsigma_m(j)$ | Age profile of SMI onset | Ext. | SMI prevalence by age, NSDUH 2024 | Profile | n.a. | Profile |
| *Block 11: Interest and taxes* | | | | | | |
| $r$ | Risk-free rate | Ext. | Kaplan et al. (2020) | 0.03 | n.a. | 0.03 |
| $\varphi$ | Tax progressivity | Ext. | Heathcote et al. (2017) | 0.85 | n.a. | 0.85 |
| $\tau_\ell$ | Level of the labour income tax | Int. | Average labour income tax, McDaniel (2007) | 22.4% | 20.9% | 0.894 |
| $\tau_k$ | Capital income tax | Ext. | US long-run rate on gains and dividends | 0.15 | n.a. | 0.15 |
| *Block 12: Social security benefits and programmes' rules* | | | | | | |
| $S(y)$ | Social security | Ext. | SSA primary insurance amount | Statute | n.a. | 0.90, 0.32, 0.15 |
| $\underline{S}$ | SSI floor over average labour income | Ext. | SSA federal benefit rate | 0.177 | n.a. | 0.177 |
| $\psi_{ui}$ | UI replacement rate | Ext. | Department of Labor, statutory rates | 0.50 | n.a. | 0.50 |
| $\bar{y}$ | Cap on the UI earnings base | Ext. | Department of Labor, state benefit maxima | 1.00 | n.a. | 1.00 |
| $\kappa_V$ | Voucher means test | Ext. | HUD rule | 0.50 | n.a. | 0.50 |
| $\kappa_r$ | Tenant's rent, share of income | Ext. | HUD rule | 0.30 | n.a. | 0.30 |
| $\varrho$ | Food stamp clawback rate | Ext. | SNAP statute, 7 CFR 273.10 | 0.30 | n.a. | 0.30 |
| *Block 13: Drug market* | | | | | | |
| $J_E$ | Maximum age at which use can begin | Ext. | Model age 55, that is age 80 | 55 | n.a. | 55 |
| $\nu_1$ | Supply shifter | Ext. | Normalisation, so $p_v = 1$ | Norm. | n.a. | Norm. |

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

[19] To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state-licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces.

**Block 7: Preferences.** The four parameters in this block say what a household wants, and what it is willing to give up to get it.

$\sigma$**, risk aversion.** It sets the marginal utility of consumption, so it affects every welfare comparison in the paper. Increase it and households want smoother consumption and insurance is worth more, so the voucher is worth more. I set it to the standard value of two, following Attanasio and Weber (2010).

$\gamma$**, the weight on housing services.** It is the weight on housing in the bundle of consumption and housing, so it sets what a dwelling is worth against consumption, and that bundle is what the drug is weighed against. As $\gamma$ rises housing is worth more against consumption, households buy more of it, and the share of spending that goes on housing rises with it. $\gamma$ is chosen for the share it produces: shelter plus utilities take 26.8 percent of expenditure in the 2024 CEX, which excludes illicit drug spending. At its value of 0.54, the model's share is 26.2 percent.

$\beta$**, the discount factor.** It sets how much households discount the future, and so how much they save and how much the prospect of death deters them from using drugs. As it rises households save more, the wealth-to-output ratio rises and households are more deterred by death. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover.

$\xi_j$**, the equivalence scale.** The household's consumption and dwelling are shared with the child while it lives at home, and thus are divided by $\xi_j$ to give what each member gets. Raise it and each member gets less of a given budget, and the child counts for more of the household, through the weight $\xi_j - 1$ in (21). It is the square-root scale, $\sqrt{N}$, where $N$ in the model is set to two while the child lives with the parent, from the parent's age 26 to 50, and to one after the child has left.

**Block 8: Rental ladder.** These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, $h_1$, which was discussed in Block 2.

$h_2$**, the smallest rental size.** The size of the cheapest dwelling is what decides how much of its income a poor household must spend to be housed at all. Raise it and the bottom rung is larger and dearer, so more households are burdened, and a voucher is worth more to them. It is chosen so that the fraction of renters paying more than half of their income for housing is similar to the data, 19.2 percent in the 2022 SCF; the model gives 25.8 percent.

$h_3/h_2$ **and** $h_4/h_2$**, the spacing of the ladder.** The second and upper rungs set how much housing they provide, and so what a household that can afford more than the smallest unit can move up to. Raise them and each step up the ladder costs more rent, and fewer households can rent them. The two larger rungs are set at 4.00 and 8.00 times the smallest.

**Block 9: Earnings and employment.** These ten parameters determine how long a household lives and works, what it is expected to earn over its life, how persistent the shocks to it are, and how it moves between work and unemployment.

$J$**,** $J_K$ **and** $J_R$**, the length of life, of childhood at home and of working life, respectively.** A household lives at most $J$ periods, has its child at home for its first $J_K$ of them and works until age $J_R$, after which it retires on social security. Together they set how many years of life and of earnings a disorder can cost, and $J_K$ how long a parent's drug use reaches its child through what they share. Raise either $J$ or $J_R$ and a disorder that shortens life or work costs more; raise $J_K$ and the budget and childhood homelessness externalities last longer. I set $J = 60$, so that with $j = 1$ at age 26 the last age is 85, $J_K = 25$, so that the child lives at home while its parent is 26 to 50, and $J_R = 41$, so that retirement begins at age 66, the full retirement age under social security for those born between 1943 and 1954.

$\varepsilon(j)$**, the age profile of efficiency.** The deterministic hump in earnings over the life cycle. Without it every household would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life. I take it as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j -$

1)/30} with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age.

$\rho_z$ **and** $\sigma_\zeta$**, the productivity process.** Log productivity is persistent but not permanent: it follows an autoregressive process with persistence $\rho_z$ and shocks of standard deviation $\sigma_\zeta$. $\sigma_\zeta$ is what decides how many households are poor enough to be at risk of homelessness, and the share of earnings that goes to the bottom half of working households falls in it. I set $\rho_z = 0.90$, the persistence of the income process in Guvenen et al. (2023). $\sigma_\zeta$ is calibrated to the bottom half's share of earnings among households aged 26 to 65 with wage income in the 2022 SCF, 17.2 percent; at $\sigma_\zeta = 0.41$ the model gives 16.2 percent.

$\rho$**, the persistence of productivity across generations.** The child's log productivity at entry into adulthood is drawn around its parent's with weight $\rho$ and around the mean with weight $1 - \rho$, so $\rho$ is how much of its parent's current productivity a child keeps. It is the channel through which anything that lowers a parent's productivity lowers its child's: a parent whose own productivity carries the penalty $\delta_d$, because its own parent had a SUD, passes a lower draw on to its child, so a parent's SUD lowers productivity beyond the next generation. Raise it and a child's prospects depend more on its parent's, and the externality grows. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005).

$\bar{\pi}^{eu}$**, the job separation rate.** The chance that an employed household loses its job each year. It is scaled by $\varpi(y)$ in (30), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it. It is chosen to reproduce the aggregate unemployment rate in the BLS CPS over 2014 to 2024, 4.9 percent, given the job-finding rate set externally below; at its value of 0.027 the model gives a 5.6 percent aggregate unemployment rate.

$\varpi(y)$**, separations by productivity.** This parameter tilts the separation rate across productivity, so that low-productivity households lose their jobs more often, which is what puts them at a higher risk of homelessness. Steepen the tilt and job loss concentrates further among the poorest, and so does the risk of homelessness. It is normalised to average one across households, so that $\bar{\pi}^{eu}$ alone sets the aggregate rate. Each productivity node takes the unemployment odds of the education group it falls in, from the BLS 2019 annual averages of unemployment by educational attainment for ages 25 and over, with education ordering productivity by median earnings.

$\bar{\pi}^{ue}$**, the annual job-finding rate.** This parameter is the probability that an unemployed household finds work each year it is unemployed. With the divisors $\kappa_d^{ue}$ and $\kappa_m^{ue}$ it sets how long an unemployed household with a SUD or SMI is expected to remain unemployed. Raise it and spells out of work are shorter, so fewer households fall behind on the rent. I set it to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

$p_{ui}$**, the share of separations that are insured.** A household that loses its job draws UI for one year with probability $p_{ui}$ and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings, and how costly being unemployed is . I set it to 0.30, since about seven in ten unemployed people who worked in the past year had not applied for UI since their last job in the BLS supplement to the CPS for 2022.

**Block 10: Baseline survival and SMI onset.** These two parameters are the survival and the SMI-onset age profiles.

$\bar{\phi}_j$**, baseline survival by age.** This is the baseline chance of surviving to the next age, before any multiplier. It is the profile every mortality multiplier in Section 5 multiplies, so it is the base the SUD, homelessness and SMI hazards scale. Raise it and households expect to live longer, so the life a SUD puts at risk is worth more. I take the profile from the life table for the total population of the United States in 2010 (Arias, 2014), for ages 26 to 85.

$\varsigma_m(j)$**, the age profile of SMI onset.** The shape of onset across ages, normalised to one at age 26, so that $\omega_m$ in Table 4 sets its level alone. It is here rather than in Table 4 because, like the age survival profile, it only sets the baseline; what actually affects the impact of Housing First versus Treatment First is $\iota$, the multiplier a SUD puts on SMI onset, which is in Table 4. It is the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024.

**Block 11: Interest and taxes.**

$r$**, the risk-free rate.** This parameter is the return on saving. Raise it and saving pays more, so households hold more assets to carry them through a spell out of work or without a voucher. I set it to three percent, from Kaplan et al. (2020).

$\varphi$**, tax progressivity.** This is the exponent on income in the tax schedule of Section 4.3, and it determines the progressivity of the labour income tax. Lower it and the tax is more progressive, so households with low incomes, among them most voucher holders and most drug users, keep more of what they earn. It is set to 0.85, which makes the schedule somewhat less progressive than the estimate of $1 - \varphi = 0.181$ in Heathcote et al. (2017).

$\tau_\ell$**, the scale of after-tax labour income.** Given $\varphi$, it sets how much of their labour income households keep after tax, so raising it lowers the tax liability. It does not move in the experiments: the budget is balanced by $\pi_V$ instead. It is calibrated to an average labour income tax rate of 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use; at its value of 0.89 the model gives 20.9 percent.

$\tau_k$**, the capital income tax.** This parameter affects the after-tax return on savings. Raise it and saving pays less, so households hold fewer assets to fall back on. I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

**Block 12: Social security benefits and programmes' rules.**

$S(y)$ **and** $\underline{S}$**, social security and its floor.** Retirement income follows the statutory primary insurance amount: it replaces a share of the household's income at its last working age that falls across the two bend points of the Social Security Administration (SSA), up to the taxable maximum, with the SSI payment as a floor. As $S(y)$ rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it. The shares $\alpha_1, \alpha_2, \alpha_3$ of the formula in footnote 11 are $0.90, 0.32, 0.15$ below the first bend point, between them and above the second, respectively. The bend points $k_1$ and $k_2$ and the taxable maximum $y^{\max}$ are the SSA's 2024 dollar amounts divided by its average wage index, 0.22, 1.33 and 2.64. The floor, $\underline{S}$, is 0.18 of average labour income.

$\psi_{ui}$ **and** $\bar{y}$**, UI.** These two set the UI replacement rate and the cap on the earnings it is paid on. Raise either and the first year out of work costs a household less, so a spell out of work is less likely to end in a shelter. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

$\kappa_V$ **and** $\kappa_r$**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's rent contribution. Raise $\kappa_V$ and more households are eligible, so more have a voucher to lose under Treatment First; raise $\kappa_r$ and a voucher pays less. I take both from HUD's own rules. I set them to half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of its income a tenant pays towards the rent, the voucher paying the rest up to the payment standard, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

$\varrho$**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. Raise it and food assistance is clawed back faster, so it supports fewer households above the very bottom. I set it to thirty percent, which is the statutory rule: a household's monthly allotment is the maximum allotment for its size reduced by thirty percent of its net monthly income (7 CFR 273.10).

**Block 13: Drug market.** The two parameters in this block describe the rest of the drug market, closing the loop on the elasticity explained in Block 6: the first fixes who can enter it, and the second is set so that supply equals demand in the benchmark, and is then held fixed.

$J_E$**, the maximum age at which use can begin.** This restricts only households that do not have a SUD, not ones that already do. It is there so that use cannot begin in the last few years of life, where almost no continuation value remains to be lost. That is an artefact of the horizon rather than a feature of the world: to keep the age dimension manageable the model imposes certain mortality after age 85, so a household near the end knows death is coming with certainty and a disorder costs it almost nothing. Without the cap the model would let use begin exactly there, and at the final age it would be worse than that, since a household that acquires the disorder in the last period dies at the end of it whatever the hazard does: the disorder carries literally no consequence and the household can only stand to benefit from the drugs. I set it to fifty-five in model terms, which corresponds to age 80 in the data, very close to the maximum age of 85.

$\nu_1$**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and kept fixed across experiments, in which the price is free. It normalises units, not behaviour.

## 5.3 Validation

Up to now, I have fitted the model to match, as closely as I could, data from the United States on the objects this paper is about. This section asks how the model performs on other relevant moments that it was not fitted to. The performance is shown in Table 6, broken into numbered blocks, with each block explaining why it matters for the experiment.

**Block 1: Homelessness.** The first row gives each age band's share of the homeless. The youngest band is the target of $T_0$; the other four are free. The model puts 37.9 percent of the homeless at ages 35 to 44, against 27.2 in HUD's PiT count, and 11.4 percent at 55 and over, against 25.9. Of the homeless in the model, 21.4 percent are employed, against 46 in the data, and their income is 6.5 percent of the average, against 19.1 in Meyer et al. (2024). For the experiment, the point is that Treatment First acts on households through the voucher they hold or lose, and its effect on children runs through the households raising them. Most of the homeless are of child-rearing age in the model, as in the data, 88.6 percent aged 25 to 54 against 74.2, so the model places the policy's margin at the ages where it reaches children.

**Block 2: Who uses and who carries a SUD.** Users in the model are somewhat younger than in the data: 67 percent of them are 26 to 34, against 45 in NSDUH 2023, and their mean age is 34.6, against 39.6 in the data. Unemployment among households with a SUD is 14.5 percent, against 16.2 in the data. Households with SMI are 2.64 times as likely as others to use drugs, against 2.44, and excess deaths due to SUD in the model account for 14.5 percent of deaths at ages 25 to 44, against 20.0 percent from drug poisonings in CDC WONDER. Treatment First and Housing First affect a child through its parent's decisions, and those who use drugs are mostly of child-rearing age, in the model as in the data: 93 percent of users are 26 to 54, against 78. And since households with SMI use drugs at higher rates than the rest of the population, the model also carries the notion that it is harder for Treatment First to discourage them from using than households without SMI.

**Block 3: Housing and vouchers.** In the model, 58.7 percent of housed households spend over thirty percent of their income on rent, against 43.2 percent of renters in the SCF. A voucher holder in the model pays 39.0 percent of its income in rent, against 31.2 in HUD's PSH, and 39.2 percent of voucher holders live mainly on wages, against 28.0. The model therefore captures how important vouchers are to households: households spend a large share of their income on housing, so without a voucher poor households have substantially less left for other consumption. The consumption that Treatment First puts at stake when it conditions the voucher is of a similar magnitude in the model as in the data.

**Block 4: Distribution.** The poorest fifth of households receive 3.40 percent of income in the model, against 2.71 in the SCF. Their share of net worth is 0.27 percent, against $-0.23$, because in the SCF the bottom of the distribution owes more than it owns, and the bottom half holds 6.3 percent of net worth,

against 2.2 in the SCF. Housing First and Treatment First act on the households that receive vouchers, and these are the poorest; the model captures their relative poverty broadly in line with the data.

**Block 5: Drug price elasticities.** The calibration targets neither the participation nor the intensive elasticity of drug demand. The model's participation elasticity is $-1.32$, somewhat larger in absolute value than the estimates of about $-0.3$ to $-0.9$ in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and its intensive elasticity, measured over all users, is $-0.63$, in the middle of the estimates of about $-0.3$ to $-0.8$ in Grossman and Chaloupka (1998) and Olmstead et al. (2015). The more elastic participation is, the more it works against Treatment First: as Treatment First lowers the drug price, more households go back to using, so its effect on drug use is smaller than it would be with a smaller participation elasticity. If Treatment First still improves welfare in the experiments below, it does so despite this channel, which makes that result conservative.

**Block 6: Transmission.** The externalities on children, which carry much of the case for either policy, run through these two links. A child of a parent with a SUD has 1.46 times the odds of acquiring one in the model, within the range of 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013). An adult who is homeless in the model has 6.23 times the odds of having had a homeless parent, above the 1.8 to 2.4 of Bassuk et al. (1997).[20] Transmission in the model is therefore at the low end of the evidence on the SUD, the margin Treatment First acts on, and above it on homelessness. Since Treatment First leaves the share of children in a shelter where it was, the excess transmission of homelessness favours neither policy, while transmission of the SUD at the low end of the evidence understates what Treatment First does for children, so if it still raises welfare in the experiments below, that result is conservative.

**Block 7: Losing a voucher.** Costantini (2025) finds that Housing First lowers the three-year mortality of homeless veterans with mental illness by 4.6 percentage points, while transitional Treatment First programmes have no lasting effect on mortality.

In the model, with the drug price and $\pi_V$ held, I compare households with SMI whom Treatment First's revocation puts on the street with households who have SMI and kept their voucher. Losing the voucher raises the former's three-year mortality by about 3.6 percentage points relative to the latter's, the same order of magnitude as his estimate. This validation matters for the experiment because a higher risk of death is the main cost Treatment First imposes on the households it screens out, and the model does not understate it.

# 6 Experiments and results

The main experiment is the reform the introduction describes. The programme stays rationed and only its rule changes, from Housing First, the benchmark, to Treatment First, under which a household that uses or carries a SUD loses its voucher if it has one, or does not receive one, when the screen detects it (Section 4.9). Three further experiments, in Section 6.2, come with it. The first is Treatment First without the externalities, with the drug price and $\pi_V$ held at their levels under Housing First. The other two give the optimal drug use, with a parent that weighs its child fully, and the optimal homelessness, with each household weighing the disamenity its own homelessness imposes on everyone else. Two objects move in the main experiment and in these last two. The probability of drawing a voucher, $\pi_V$, adjusts so that net government revenue stays at its benchmark level, $\bar{R}$. The drug price clears the drug market, and the tax system is held where it is in the benchmark.

**The screen.** Under Treatment First a household fails the condition if it uses drugs or carries the disorder. A holder that fails is caught with probability $p_{\text{det}}$ and loses its voucher, and an applicant that draws a

[20] Bassuk et al. (1997) compare 220 homeless mothers with 216 low-income housed mothers, so both groups are poor and the odds ratios, 1.8 for a homeless female caretaker and 2.4 for a male one, separate the transmission of homelessness from the poverty that accompanies it. A ratio against the general population would also credit the housing penalty with everything else that differs between poor children.

Table 6: Untargeted moments

| Moment | Data | Model | Source |
|---|---|---|---|
| *Block 1: Homelessness* | | | |
| Share of homeless by age, 25–34 to 65+[a] | 26.0, 27.2, 21.0, 18.4, 7.5 | 26.4, 37.9, 24.2, 10.7, 0.8 | HUD PiT 2024 |
| Employed share of the homeless | 46 | 21.4 | Meyer et al. (2024) |
| Income of the homeless, over average | 19.1 | 6.5 | Meyer et al. (2024) |
| *Block 2: Who uses and who carries a SUD* | | | |
| Share of users by age, 26–34 to 65+ | 45, 19, 14, 12, 10 | 67, 16, 10, 5, 2 | NSDUH 2023 |
| Mean age of users, years | 39.6 | 34.6 | NSDUH 2023 |
| Unemployment given a SUD | 16.2 | 14.5 | NSDUH 2024 |
| Drug use with SMI over without, ratio | 2.44 | 2.64 | NSDUH 2024 |
| Drug deaths as a share of deaths, 25–44 | 20.0 | 14.5 | CDC WONDER |
| *Block 3: Housing and vouchers* | | | |
| Renters paying over 30% of income | 43.2 | 58.7 | SCF 2022 |
| Rent over income, voucher holders | 31.2 | 39.0 | HUD PSH 2023 |
| Voucher holders living on wages | 28.0 | 39.2 | HUD PSH 2023 |
| *Block 4: Distribution, shares of the total* | | | |
| Income, bottom 1, 5, 10, 20% | 0.01, 0.31, 0.92, 2.71 | 0.01, 0.19, 1.00, 3.40 | SCF 2022 |
| Net worth, bottom 1, 5, 10, 20, 50% | −0.16, −0.28, −0.29, −0.23, 2.2 | 0.00, 0.00, 0.00, 0.27, 6.3 | SCF 2022 |
| *Block 5: Drug price elasticities* | | | |
| Participation (extensive margin) | −0.3 to −0.9 | −1.32 | Saffer and Chaloupka (1999), Grossman and Chaloupka (1998) |
| Intensive margin (users) | −0.3 to −0.8 | −0.63 | Grossman and Chaloupka (1998), Olmstead et al. (2015) |
| *Block 6: Transmission, odds ratios* | | | |
| Child of a SUD parent acquires a SUD | 1.44, 2.51 | 1.46 | Mellentin et al. (2016), Yoon et al. (2013) |
| Homeless adult had a homeless parent | 1.8 to 2.4 | 6.23 | Bassuk et al. (1997) |
| *Block 7: Losing a voucher* | | | |
| Three-year mortality, households with SMI put on the street, pp[b] | 4.6 | 3.6 | Costantini (2025) |

Notes: percent, except the elasticities and the odds ratios of the transmission block, and percentage points in Block 7; the model is the Housing First benchmark. [a]The share aged 25 to 34 is the target of $T_0$ in Table 4, so of this row only the other four bands are untargeted. [b]Data: Costantini's instrumental-variables estimate of how much enrolling in the Housing First programme of the Department of Veterans Affairs lowers the three-year mortality of homeless veterans with mental illness. Model: the three-year mortality gap between the households with SMI whom Treatment First's revocation forces into homelessness and the same households, also with SMI, who kept their voucher, with the drug price and $\pi_V$ held.

voucher is screened the same way before it receives one. A household that is caught is excluded from the programme whatever its income, and returns to the queue for a voucher with probability $\psi_x$ a year. The government pays $c_{\text{test}}$ for every household it screens, inside the budget that $\pi_V$ balances. None of these three parameters does anything in the benchmark, where no voucher is conditional, so they are set here rather than in Section 5.

I set the detection rate to $p_{\text{det}} = 85$ percent. In a blinded study of 1,000 pain patients, office urine screens detected 92.5 percent of opioid use, against confirmation by liquid chromatography and tandem mass spectrometry (Manchikanti et al., 2011). Laboratory screens detect 96 to 100 percent across five opioid assays (Walsh et al., 2025). The rate in the model is below these, which is conservative, since a lower rate weakens Treatment First. A household that neither uses nor carries the disorder is never flagged. A screen is positive only above a cutoff concentration, and a positive screen is confirmed in a laboratory before it has consequences, so a false positive is all but ruled out.

The release rate is $\psi_x = 1/3$, an expected exclusion of three years. That is the same period for which the Housing Choice Voucher programme already bars a household evicted from federally assisted housing for drug-related criminal activity (Code of Federal Regulations, Title 24, Section 982.553). The test costs $c_{\text{test}} = 0.031$ percent of average earnings in the economy, or \$35 a test. That is the upper end of the estimates: the Texas Legislative Budget Board's fiscal note for Senate Bill 11 puts a test at \$35 (Legislative Budget Board, 2013), which again loads the experiment against Treatment First.

## 6.1 The welfare criterion

Every welfare number in this paper is a CEV, computed as follows. Write $V^b$ for lifetime value under the benchmark and $V^c$ for lifetime value under a counterfactual policy. The CEV $\Delta$ is the proportional change in consumption, applied at every date and in every state, that makes a newborn under the benchmark as well off as the same newborn under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{40}$$

A positive $\Delta$ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

**Whose welfare.** Welfare here is measured for newborns. It includes childhood in the parent's household and adult life after it. The childhood part is the child's own flow utility from the bundle it shares with its parent. It is counted as $\xi - 1$ of the household, as it is in the parent's problem, and with the flow value of being alive, $(\xi - 1)\big[U(X_c) + \aleph\big]$, less the disamenity of aggregate homelessness $\kappa_x H$ that the child bears like everyone else. This flow is averaged at each age over the households of that age, summed over the years the child lives at home and discounted to its birth. The adult part is the child's value upon entering the economy at age 26, discounted by the same years. The child's own childhood counts at weight one here, not at its parent's weight $\omega_c$, and that is what brings the externality on the child into welfare. Its adult part is the value its own problem maximises, so the child it raises in turn enters there at the weight $\omega_c$.

The value is taken before any of its draws are realised, including the household it is born into, so $V^b$ and $V^c$ are expectations over the productivity draw and everything that follows from it, and $\Delta$ equates them. The expectation is taken under each policy's own equilibrium distribution rather than a common one. The penalties that set a child's starting productivity depend on what happened to its parent, so the distribution a newborn is drawn from is itself something the policy moves. The ex-ante expected newborn welfare is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023).

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state. Because both are the ex-ante value of a child at birth, childhood included, any difference between them is the transition itself rather than a change in who is being counted.

## 6.2 The long run

Table 7 sets the long run under Treatment First and the two optimal experiments beside the benchmark, and, in the first two columns, Housing First and Treatment First without the externalities.

**Treatment First without externalities.** Without the externalities, the full model gives the ranking of the stylized model. To capture the no-externality case, I set the aggregate homelessness externality to zero, $\kappa_x = 0$, and treat the child as appearing only as an adult: its parent does not weigh it, its productivity does not depend on what its parent did, and its childhood, the years its parent is 26 to 50, is not counted in welfare. With the drug price and $\pi_V$ held at their levels under Housing First, Treatment First cuts drug use by 60 percent and the share carrying a SUD by 61 percent, against Housing First with the same externalities switched off, while homelessness rises by 3 percent (first two columns of Table 7). And yet, no household is better off under Treatment First, with newborns losing 1.50 percent of lifetime consumption as adults. What follows therefore rests on the externalities on children.

**Treatment First.** Moving to full general equilibrium, with an endogenous drug price and the government's budget clearing, Treatment First cuts drug use by 32 percent, from 8.96 to 6.09 percent of households, and the share of households carrying a SUD by 33 percent, from 3.81 to 2.54 percent of households. It also raises the share of households with a SUD who are in treatment, from 24.9 to 29.3 percent. Treatment only buys a higher chance of recovering, and Treatment First makes recovering worth more: a household that has recovered and does not use passes the screen and can hold a voucher if it is eligible and offered one. As users who lose their vouchers leave the programme, it costs less at a given $\pi_V$. Tax revenue also rises, as fewer adults remain unemployed because of a SUD and more children inherit higher productivity draws. Thus, $\pi_V$ rises from 6.75 to 9.62 percent, by 42 percent, making staying clean of drugs even more desirable. At the same time, taking users out of the drug market lowers the drug price by 14 percent, which draws some households back.[21] Had the price stayed at its benchmark level, drug use would have fallen by 74 percent instead.

Treatment First also lowers homelessness by 1 percent, from 0.62 to 0.61 percent of households, as the balance of three effects. First, it revokes the vouchers of households with a SUD and puts them at a higher risk of homelessness. Of them, 5.63 percent are homeless, against 3.28 under Housing First. On its own this raises homelessness by 10 percent. Second, less SUD means less unemployment and so less homelessness.

[21] The qualitative results continue to hold with a detection rate of 70 percent and with a more elastic supply of the drug.

There are a third fewer households with a SUD, which on its own lowers homelessness by 6 percent. Third, the vouchers it revokes go, through the higher $\pi_V$, to other clean and eligible households, lowering homelessness by 5 percent. Together the last two more than offset the first. For households aged 26 to 50, the years a child lives with them, the same three effects nearly net to zero, so the share of children in a shelter remains unchanged at 1.03 percent. These are long-run outcomes. However, in the first years after the reform both homelessness and childhood homelessness rise, because the households that already carry a SUD lose their vouchers at once, while their number falls only as fast as recovery allows (Section 6.3).

A newborn is better off under Treatment First by 4.33 percent of lifetime consumption. Welfare rises because the cut in drug use reduces the budget and earnings externalities and, through the government budget, pays for a larger voucher programme, while homelessness, and with it the externalities that run through it, is almost unchanged in the long run. Without the larger voucher programme, the reform would lose nearly all of its gain: with $\pi_V$ held at its benchmark level, so that the revenue it frees goes unspent, a newborn would gain 0.18 percent. The cheaper drug, by contrast, works against the overall gain, since it draws users back: with the drug price held at its benchmark level and $\pi_V$ still balancing the budget, a newborn would gain 13.11 percent. With both held at their benchmark values, so that only the screen acts, a newborn would gain 5.04 percent: the screen on its own delivers more than the whole reform, the cheaper drug it brings about takes back nearly all of that, and recycling the revenue into vouchers restores most of what is lost.

About 57 percent of the gain in lifetime utility comes in childhood. Parents spend more on the consumption and housing they share with the child and less on the drug, and they are less likely to acquire a SUD and remain unemployed longer because of it, raising childhood consumption further. The share of children living with a parent who uses drugs falls from 16.32 to 10.99 percent, and the share living with one who carries a SUD falls from 5.81 to 3.89 percent.

The other 43 percent comes in adult life, from two sources. The first, 11 percent of the whole gain, is the better productivity a child draws when its own parent does not carry the disorder. Escaping the penalty is worth a lot to each child it spares, about 12 times what the average child gains at the same productivity. But it spares only about 3 percent of children: the share of newborns who enter adult life with the penalty falls from 8.99 to 6.07 percent.

The second, the remaining 32 percent, does not come from the child's productivity at all. It is a result of the better economy it lives its adult life in. A clean and eligible household is 42 percent more likely to draw a voucher. A voucher is worth most to a household that enters adult life with low productivity, because it is the likeliest to need one, and the value of entering adult life rises about 30 times as much at the lowest productivity as at the highest. This part of the gain is therefore larger for a child whose own parent carried the disorder, since the penalty leaves it with lower productivity. Moreover, even a child who goes on to use drugs pays less for them: the drug is 14 percent cheaper in equilibrium, so it can consume more of it. Thus, every household that enters adult life at 26 is better off under Treatment First than it would be in the same state under Housing First (Table 7). Of newborns, 92 percent are better off under Treatment First than they would be if born into the same household under Housing First. Across all households, 76 percent are better off, but only 20 percent of those that carry a SUD, since they lose access to vouchers.

**Optimal drug use.** The third column keeps Housing First but has each parent weigh its child as itself, $\omega_c = 1$: the drug use parents would choose if they took in full what their use does to their children. A parent that counts its child's consumption as its own uses less of the drug, and some parents stop altogether. What it spends on the drug (and any earnings it loses while it remains unemployed longer because of a SUD) comes out of the bundle it shares with the child, and a SUD penalises the child's earnings. But, because supply is not perfectly elastic, the fall in demand then lowers the drug price, which draws some of them back.

Notice that Treatment First cuts drug use by more than full altruism would, 32 percent against 17, but homelessness by less. Under full altruism homelessness falls by 9 percent, and the share of children in a shelter falls by 14 percent. Treatment First, by contrast, lowers homelessness by only 1 percent and leaves the share in a shelter where it was. Homelessness falls because a parent that counts its child as itself bears in full what a shelter costs the child, and spends what it has to keep the child housed. Thus, the fall comes in the years a child lives at home, whereas among households aged 51 to 64, whose children have left, homelessness rises by 10 percent.

Table 7: The long run under each policy

| | No externalities | | | | | |
|---|---|---|---|---|---|---|
| | Housing First | Treatment First | Housing First | Treatment First | Optimal drug use | Optimal homelessness |
| *Adults* | | | | | | |
| Drug price, benchmark = 1 | 1.03 | 1.03 | 1.00 | 0.86 | 0.88 | 0.88 |
| Voucher probability $\pi_V$ | 6.75 | 6.75 | 6.75 | 9.62 | 5.84 | 5.85 |
| Drug use | 9.01 | 3.59 | 8.96 | 6.09 | 7.45 | 7.44 |
| Carrying a SUD | 3.84 | 1.51 | 3.81 | 2.54 | 3.13 | 3.13 |
| Homelessness | 0.60 | 0.62 | 0.62 | 0.61 | 0.56 | 0.55 |
| SUD share of the homeless | 20.5 | 16.1 | 20.2 | 23.3 | 13.3 | 13.3 |
| In treatment, of those with a SUD | 25.0 | 29.9 | 24.9 | 29.3 | 25.2 | 25.2 |
| *Children* | | | | | | |
| Living with a parent who uses drugs | 16.42 | 6.57 | 16.32 | 10.99 | 13.48 | 13.47 |
| Living with a parent who carries a SUD | 5.84 | 2.32 | 5.81 | 3.89 | 4.78 | 4.78 |
| In a shelter | 1.01 | 1.04 | 1.03 | 1.03 | 0.89 | 0.87 |
| *Welfare* | | | | | | |
| CEV of a newborn, percent of lifetime consumption | – | −1.50 | – | 4.33 | – | – |
| Share of the newborn's gain that comes in childhood | – | – | – | 57 | – | – |
| Households entering adult life at 26 better off | – | 0 | – | 100 | – | – |
| All households better off | – | 0 | – | 76 | – | – |
| Households with a SUD better off | – | 0 | – | 20 | – | – |
| Newborns better off, by the household they are born into | – | – | – | 92 | – | – |

Notes: percent unless the row says otherwise; each column is a steady state. The first two columns are Housing First and Treatment First without the externalities, with $\kappa_x = 0$ and the child appearing only as an adult. Under Housing First the drug price clears and $\pi_V$ is the benchmark's; under Treatment First both are held there. The CEV in the second column is the newborn's value at 26, against the first column. Housing First, the third column, is the benchmark. Treatment First screens with a detection rate of 85 percent, and $\pi_V$ holds net government revenue at its benchmark level in every column but the first two. Optimal drug use is Housing First with $\omega_c = 1$, and optimal homelessness adds to optimal drug use each household weighing the disamenity of its own homelessness. Children are those living with households aged 26 to 50, the years a child lives at home. The CEV is against the benchmark. A household is better off when its value under Treatment First exceeds its value in the same state under Housing First, using the Housing First distribution.

$\pi_V$ falls from 6.75 to 5.84 percent to hold net government revenue at its benchmark level. A population that uses fewer drugs lives longer, so the government pays more in social security and each voucher holder holds its voucher for longer. Hence, at the benchmark's $\pi_V$ the programme would cost 5 percent more. The same happens under Treatment First, but there the users the screen takes out of the programme outweigh it, and at the benchmark's $\pi_V$ the programme would cost 22 percent less. I do not report a CEV here because the weight on the child changes the welfare function itself.

**Optimal homelessness.** The last column is the same as the optimal drug use case, with the addition of each household weighing the disamenity its own homelessness imposes on everyone else. It reduces homelessness by only 2 percent on top of what optimal drug use achieves, which shows that this externality turns out to be very small. Households can save and use less of the drug to improve their housing outcomes, and many do when they weigh their child, as the optimal drug use column shows. Nevertheless, the disamenity their homelessness imposes on others is not a strong enough deterrent, so weighing it does not alter their behaviour significantly. $\pi_V$ remains practically unchanged. Again, I report no CEV here because it too changes the welfare function.

## 6.3 The transition

**[UPDATE: transition: provisional, the second iteration of the new sixty-year run, which has not yet converged; every number and figure here is refreshed as it converges]** Up until this point, I have discussed the implications of Treatment First across various dimensions in the long run. However, a major consideration for any policy change is its short-run impact on those dimensions as well. To evaluate those impacts, I compute the transition from Housing First to Treatment First. The transition starts from the benchmark, with the reform landing as an unexpected permanent shock at the start of the first year, and runs sixty years to the Treatment First steady state. The drug market clears in every year, and the budget in every year after the first, whose spending is set by the vouchers households already hold. Figure 3 shows

UPDATE

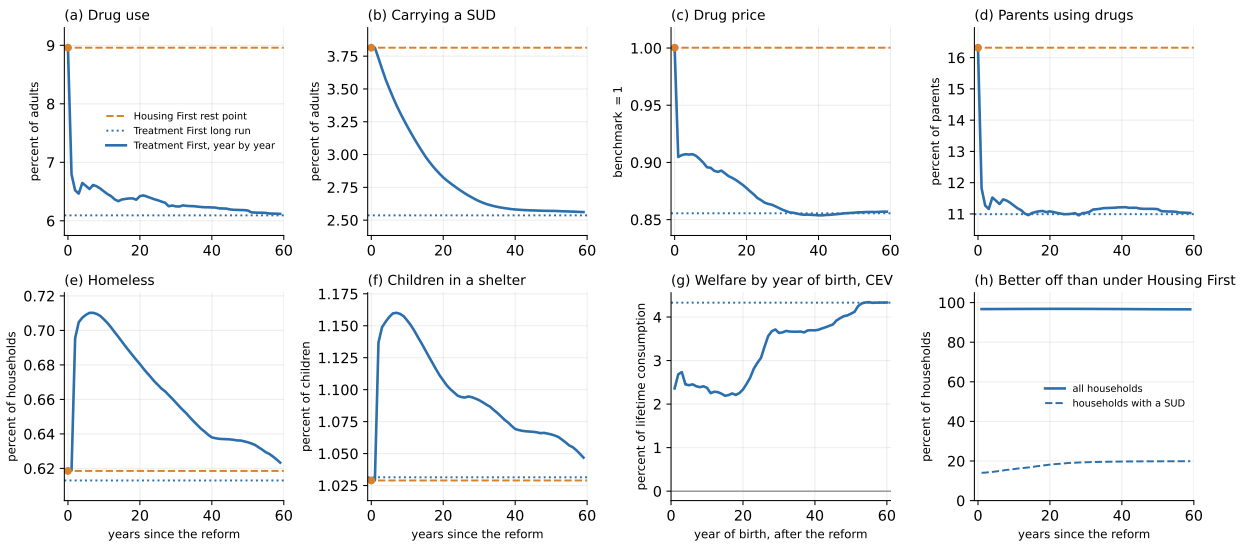



Figure 3: The transition from Housing First to Treatment First, quantitative model

Notes: panels (a) to (f) and (h) are by year since the reform, which lands at the start of year 1, with the benchmark as year 0; the dashed orange line is the Housing First benchmark, the dotted blue line the Treatment First long run and the solid blue line the path. Children are those of households aged 26 to 50. Panel (g) is the CEV of each cohort born on the path, against the benchmark, by its year of birth. Panel (h) is the share of households better off in each year of the path than in the same state under Housing First, weighted by the benchmark's distribution.

what the transition looks like along various key dimensions.

Panels (a) and (d) show drug use and the share of parents who use. Drug use falls at once: 76 percent of its long-run fall happens in the first year. This is because the screen acts on the decision to use from the day the reform lands. A household that uses risks losing its voucher at once, or having a voucher offer revoked later, when it may need one but carries a SUD, while a disorder takes years to clear. The share of parents using drugs falls from 16.32 to 11.82 percent, above its long-run level of 10.99 percent. Panel (b) shows the share carrying a SUD. This share falls slowly, because recovery is slow: half of its long-run fall is done by year 11, and nine tenths by year 29. That stock keeps demand for the drug up, and the drug price stays above its long-run level for 33 years (panel (c)). The elevated price deters young adults who have not yet started: in the first decade the number of drug users without a SUD falls by up to 46 percent, against 31 percent in the long run.

Panels (e) and (f) show homelessness and the share of children in a shelter. Homelessness overshoots. It rises by 15 percent at its peak, in year 7, and then declines slowly towards its long-run level, 1 percent below that under Housing First. A similar pattern is observed for childhood homelessness: the share of children in a shelter rises from 1.03 to 1.16 percent in year 7, against 1.03 percent in the long run.

Panel (g) shows the CEV of each cohort by its year of birth. Children born at the reform gain about 2.4 percent of lifetime consumption, less than those born in the long run. They gain, because the fall in parental drug use reaches far more children than the overshoot in homelessness: in the first year the share of parents who use drugs falls by 4.50 percentage points, while the share of children in a shelter rises by at most 0.13 points, so the first reaches about 35 times as many children as the second. They gain less than in the long run, because they grow up while the share of parents carrying a SUD is still close to its Housing First level, since that share falls only as fast as recovery allows, so more of them spend their childhood with a parent who carries the disorder, and more enter adult life with the earnings penalty. The gain stays near that level for children born while that share is still falling, and rises toward the long run for those born later, whose parents were themselves born after the reform. Children born in the first ten years gain 2.47 percent on average. Of the adults alive at the reform, 97 percent are better off under it, and even 14 percent of those who carry a SUD when it lands benefit, because the lower drug price more than offsets the loss from the sobriety requirement for some of them. Panel (h) repeats the comparison in every year of the path, for

a household in each state weighted as under Housing First. The share better off stays at about 97 percent, and the share among those with a SUD rises from 14 to 20 percent as the price comes down. Every child and household entering adult life is better off throughout.

# 7 Discussion and implications

Should housing assistance require sobriety? When households are forward looking and weigh the true risk of a SUD, the moral hazard that dominates the debate does not on its own justify conditionality: without externalities Housing First would win. What turns the ranking are externalities imposed on children, together with what removing them does to the government budget and the drug market. In the long run of a general-equilibrium model calibrated to United States data, Treatment First cuts drug use by 32 percent and the share of households with a SUD by 33 percent, and actually lowers homelessness by 1 percent. As a result, it raises the welfare of a newborn by 4.33 percent of lifetime consumption. Most of that gain comes in childhood, from parents who spend less on drugs and are less likely to remain unemployed because of a SUD after losing their job. The rest comes from better earnings and from a larger voucher programme, which the users it excludes and the higher revenues from the taxes paid by more productive and more employed households pay for.

Children born at the reform gain less than those born in the long run, because they grow up while the share of parents carrying a SUD is still close to its Housing First level, but they still gain. Yet, from a policy point of view the transition can be deceptive. The households that carry a SUD when the reform lands lose their vouchers at once, so homelessness first rises, by up to 15 percent, and the share of children in a shelter with it. It comes down only as the number with a SUD falls, which takes 29 years to complete nine tenths of its fall. And of those carrying a SUD at the reform, only 14 percent are better off.

Three lessons follow. First, the arguments for Housing First and Treatment First made by different advocacy groups rest on the wrong grounds, and the reform will thus be judged on the wrong outcomes. Treatment First supporters argue moral hazard, which in this model is not a reason for it, while its critics count homelessness, which Treatment First does not raise in the long run. The key, which neither side counts, is the impact on the children of the households it deters from using, in what those children consume while at home and in what they earn as adults; those are the outcomes on which it should be judged.

Second, the politics of the reform run against its own payoff. Its apparent costs come first, falling on households already carrying a SUD, and appear in the homelessness counts during the transition from the Housing First benchmark to the long-run equilibrium under Treatment First; its positive effects on homelessness only come later. Judged on its first years, by the PiT count, the reform will look like a failure exactly when the model's reading is that it is on course, and it will be most exposed to reversal before its long-run positive effects on homelessness, the SUD and, most importantly, children have arrived in full.

Third, the results point to two choices of design that the current debate leaves out. The first is what to do with the excess funds the government collects under Treatment First. In my experiment, all of them go back into vouchers, so an eligible household that stays clean is 42 percent more likely to get one, which is one of the reasons a newborn's adult life is worth more, and homelessness actually declines slightly in the long run, with childhood homelessness unaffected. A sobriety requirement introduced without recycling that excess revenue back into more vouchers would give up nearly all of the gain: with $\pi_V$ held at its benchmark level, a newborn would gain 0.18 percent rather than 4.33, and homelessness would rise by 13 percent.

The second is whom the condition applies to, which may further mitigate the impact on the households that already carry a SUD along the transition. The gain comes from deterrence: adults who have not yet started using stay clean to keep their access to a voucher. The early cost falls on a different group, the households that already carry a SUD when the reform lands. They cannot respond quickly, because recovery from a SUD is slow, so for them the condition brings mostly the cost. Their homelessness rises at once, and only 14 percent of them are better off. A condition applied only to new voucher holders, with current holders given a five-to-ten-year exemption, for example, may strike a good balance between keeping the threat for those who can still respond to it and sparing those who cannot.

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